Algebraic geometry 2026

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Chapter 1

Mynotes from 2026 class

Some references used here:

- Mumford's red book [3]
- Humphreys Linear algebraic groups [2]

1.1 Lecture 1 and 2

First some definitions and lemmas worth to remember.

- (Integral extension). Let R be a ring and $S \subset R$ a subring. We say that R is integrally dependent over S if every $r \in R$ is integral over S, that is, there exist a monic polynomial $f \in S[X]$ such that $f(r) = 0$.

  Lemma 1.1.1. Let $S \subset R$ be rings. A finite subset of elements $b_1, \ldots, b_m \in R$ are all integral over S if and only if $S[b_1, \ldots, b_m]$ is finitely generated (as a $S-$module) over S.

  Proof: ($\Rightarrow$) Let $b \in R$ be integral over S, i.e., there exist

  $$f = X^d + a_1 X^{d-1} + \cdots + a_d \in S[X] \tag{1.1}$$

  such that $f(b) = 0$. Then, using long division, we see that $S[b] = S + \cdots + Sb^{d-1}$ which is finitely generated. Now, if $b_1, \ldots, b_m$ are all integral over S, we can argue by induction that $S[b_1, \ldots, b_{m-1}]$ is finitely generated over S. However, since b_m is integral over $S[b_1, \ldots, b_{m-1}]$, $S[b_1, \ldots, b_m]$ is finitely generated over $[b_1, \ldots, b_{m-1}]$ and it must be finitely generated over S aswell.

  ($\Leftarrow$) Suppose that $S[b_1, \ldots, b_m] = S\beta_1 + \cdots + S\beta_n$ for some $\beta_i \in R$. Then for any $b \in R$ which fixes $S[b_1, \ldots, b_m]$, we have $b\beta_i = \sum_j a_{ij}\beta_j$ for some $(a_{ij}) \in \mathrm{Mat}_n(S)$. Therefore

  $$0 = (bI - (a_{ij}))^*(bI - (a_{ij}))v = \det(bI - (a_{ij}))v \tag{1.2}$$

where $v = (\beta_1, \ldots, \beta_n)^T$. Therefore $\det(bI - (a_{ij}))\beta_i = 0, \forall i$ and since $1 \in S\beta_1 + \cdots + S\beta_n$, we see that $\det(bI - (a_{ij})) = 0$. Since $\det(X - (a_{ij})) \in S[X]$ is monic, we are done. □

Corollary 1.1.2. (Transitive property of integral dependence). Given rings $R \supset S \supset T$, if R is integral over S and S is integral over T, then R is integral over T.

Question: Do we need Noetherian hypothesis using this statement?

Proof: Let $r \in R$, then by hypothesis there exist $f = X^d + s_1 X^{d-1} + \cdots \in S[X]$ such that $f(r) = 0$. Define $A = T[s_1, \ldots, s_d]$, then $A[r]$ is a finitely generated $A-$module because $S \subset A$ and lemma (1.1.1). Now, since A is a finitely generated $T-$module by lemma (1.1.1), we see that $A[r]$ is a finitely generated $T-$module. Now, assuming T is Noetherian we see that $T[r]$ is finitely generated as a $T-$module. □

Remark 1.1.3. We can certainly say (using the previous proof) that if $r \in R$, then $S[r]$ is finitely generated as a $T-$module.

Lemma 1.1.4. Let $K \supset k$ (k a perfect field) be a finitely generated extension field with transcendence degree $\partial_k(K) = d$, then there exist generators $z_1, \ldots, z_d, z_{d+1}$ such that

(1) $K = k(z_1, \ldots, z_{d+1})$
(2) $\{z_1, \ldots, z_d\}$ are algebraicaly independent
(3) z_{d+1} is separable over $k(z_1, \ldots, z_d)$

Lemma 1.1.5. (Noether's normalization lemma - Mumford's red book). Let R be a finitely generated algebra over k. Assume also that R is an integral domain. If R has transcende degree n over k (which means that $\partial_k \mathrm{Frac}(R) = n$), then there exist $x_1, \ldots, x_n \in R$ algebraicaly independent and such that R is integral over $k[x_1, \ldots, x_n]$.

TODO: Study prove this version of the lemma using the statement made in class.

Lemma 1.1.6. (Noether's normalization lemma - Statement made in class). Let R be a finitely generated $k-$algebra. Then there exist $x_1, \ldots, x_n \in R$ algebraically independent over k such that R is integral over $k[x_1, \ldots, x_n]$.

Proof (Nagata): Since R is finitely generated as a k-algebra, there exist generators $y_1, \ldots, y_m$ such that $R = k[y_1, \ldots, y_m]$. Now suppose that $y_1, \ldots, y_m$ are algebraically independent, then we can take $x_i = y_i$. If not, then there exist $p \in k[Y_1, \ldots, Y_m] \setminus 0$ such that $p(y_1, \ldots, y_m) = 0$. Write

$$p(Y_1, \ldots, Y_m) = \sum_{\alpha \in \mathbb{Z}^m_{\geq 0}} a_\alpha Y_1^{\alpha_1} \cdots Y_m^{\alpha_m} \tag{1.3}$$

and define a new polynomial

$$\begin{aligned} q(Y_1, \dots, Y_m) &= p(Y_1, Y_2 + Y_1^{r_2}, \dots, Y_m + Y_1^{r_m}) \\ &= \sum_{\alpha} a_\alpha (Y_1^{\alpha_1 + r_2\alpha_2 + \cdots + r_m\alpha_m} + \cdots) \end{aligned} \tag{1.4}$$

for some choice of positive integers $r_2, \dots, r_m$. Take a positive number $b > \alpha_i,\ \forall \alpha_i$ such that $a_\alpha \neq 0$. If we take $r_i = b^i$, then $\alpha_1 + \alpha_2 b + \cdots + \alpha_m b^{m-1}$ are all different numbers in base b. So, from different powers of the form $\alpha_1 + r_2\alpha_2 + \cdots + r_m\alpha_m$ we have a maximal one $d = \max\{\alpha_1 + \cdots r_m\alpha_m : a_\alpha \neq 0\}$ and

$$q(Y_1, \dots, Y_m) = aY_1^d + \text{smaller degree terms} \in k[Y_2, \dots, Y_m][Y_1],\ a \neq 0. \tag{1.5}$$

Now, if we let $z_i = y_i - y_1^{r_i} \in R, i = 2, \dots, m$, we see that $q(y_1, z_2, \dots, z_m) = p(y_1, \dots, y_m) = 0$ and y_1 is integral over $k[z_2, \dots, z_m]$. By induction on number of generators, there exist $x_1, \dots, x_n$ algebraically independent such that $k[z_2, \dots, z_m]$ is integral over $k[x_1, \dots, x_n]$. So, by lemma 1.1.1 and its corollary (1.1.1), we see that $k[y_1, z_2, \dots, z_m]$ is integral over $k[x_1, \dots, x_m]$. Now, by definition, it's clear that $k[y_1, z_2, \dots, z_m]$ is a finitely generated $R-$module. So by the corollary (1.1.2) again, we see that R is integral over $k[x_1, \dots, x_m]$. □

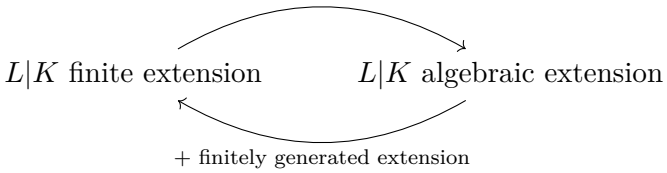
Lemma 1.1.7. Let $S \subset R$ be a ring extension. If R is a field, which integral over S, then S is a field.

Proof: Let $s \in S \setminus 0$ and $r \in R$ be its inverse. Since R is integral over S, there exist $f = X^d + a_1X^{d-1} + \cdots \in S[X]$ be such that $f(r) = 0$. Therefore

$$0 = s^{d-1}f(r) = r + a_1 + sa_2 + \cdots + s^{d-1}a_0 \Rightarrow r \in S. \tag{1.6}$$

□

Remark 1.1.8. Let $L|K$ be a field extension. Then



Theorem 1.1.9. (Zariski). Let k be a field and $\mathfrak{m} \subset R = k[X_1, \dots, X_m]$ be a maximal ideal. Then $L = R/\mathfrak{m}$ is a finite extension field of k.

Proof: By Noether normalization lemma (1.1.6), there exist $a_1, \dots, a_n \in L$ algebraically independent such that L is integral over $k[a_1, \dots, a_n]$. By lemma (1.1.7), we conclude

that $k[a_1, \ldots, a_n]$ is a field. Suppose that $n > 0$, so a_1 has an inverse. This means that $a_1 g(a_1, \ldots, a_n) = 1$ for some $g \in k[X_1, \ldots, X_n]$, which constradicts the fact that $a_1, \ldots, a_n$ are algebraically independent. Then we conclude that L is an integral (and then algebraic) over k. However, since L is a finitely generated extension ($L = k(\overline{X}_1, \ldots, \overline{X}_m)$) over k we conclude that L is in fact a finite field extension over k. □

Theorem 1.1.10. Suppose that k is an algebraically closed field. Let $I \subset R = k[X_1, \ldots, X_m]$ be an ideal, then $V(I) = \varnothing \iff I = k[X_1, \ldots, X_m]$.

Proof: If $I \neq R$, then there exist a maximal ideal $\mathfrak{m} \supset I$. So $V(I) \supset V(\mathfrak{m})$ and it's sufficient to prove that $V(\mathfrak{m}) \neq \varnothing$. By theorem (1.1.9), we have that $L = R/\mathfrak{m}$ is a finite $k-$extension, so we must have $L = k$ because k is algebraically closed. So there exist $a_i \in k, i = 1, \ldots, m$ such that $X_i - a_i \in \mathfrak{m}$ and $(X_1 - a_1, \ldots, X_m - a_m) \subset \mathfrak{m}$. However $\varphi : R \to k : F \mapsto F(a_1, \ldots, a_m)$ has kernel $\mathrm{Ker}(\varphi) = (X_1 - a_1, \ldots, X_m - a_m)$ and since k is a field, $R/\mathrm{Ker}(\varphi)$ must be a field and $\mathrm{Ker}(\varphi)$ is a maximal ideal. Therefore we conclude that $\mathfrak{m} = (X_1 - a_1, \ldots, X_m - a_m)$. Finally $V(X_1 - a_1, \ldots, X_m - a_m) = \{a_1, \ldots, a_m\} \neq \varnothing$. □

Remark 1.1.11. The hypothesis $k = \overline{k}$ in the previous theorem is crucial. For example if $k = \mathbb{R}$, we have $V(x^2 + 1) = \varnothing$.

Corollary 1.1.12. From the proof of the previous theorem, we see that there exist a bijection

$$\begin{aligned} k^n &\longleftrightarrow \left\{ \begin{matrix} \text{maximal ideals} \\ \mathfrak{m} \subset R \end{matrix} \right\} \\ (a_1, \ldots, a_n) &\longmapsto (X_1 - a_1, \ldots, X_n - a_n) \end{aligned} \tag{1.7}$$

Theorem 1.1.13. (Hilbert Nullstelensatz). Let $A \subset k[X_1, \ldots, X_n]$ be an ideal, then

$$IV(A) = \sqrt{A}. \tag{1.8}$$

Proof: The inclusion $\sqrt{A} \subset IV(A)$ is easy. Now, assume $g \in IV(A)$, where $A = (f_1, \ldots, f_m)$. This means that we have the hypothesis:

$$f_i(a_1, \ldots, a_n) = 0, \ \forall i = 1, \ldots, m \Rightarrow g(a_1, \ldots, a_n) = 0 \ (*) \tag{1.9}$$

Now, define the ideal

$$I = Ak[X_1, \ldots, X_{n+1}] + (1 - g(X_1, \ldots, X_n)X_{n+1}). \tag{1.10}$$

If I is a proper ideal, then by corollary (1.1.12), we have $I \subset (X_1 - a_1, \ldots, X_{n+1} - a_{n+1})$, which is the kernel of the evaluation map $k[X_1, \ldots, X_{n+1}] \to k : f \mapsto f(a_1, \ldots, a_{n+1})$. This means that

$$\begin{cases} f_i(a_1, \ldots, a_n) = 0, \ \forall i = 1, \ldots, m \\ g(a_1, \ldots, a_n)a_{n+1} = 1 \end{cases} \tag{1.11}$$

which contradicts our initial hypothesis $(*)$. Therefore, I is not a proper ideal, so $1 \in I$ and there exists $h_1, \ldots, h_{m+1} \in k[X_1, \ldots, X_{n+1}]$ such that

$$1 = \sum_{j=1}^{m} h_j(X_1, \ldots, X_{n+1}) f_j(X_1, \ldots, X_n) + h_{m+1}(X_1, \ldots, X_{n+1})(1 - g(X_1, \ldots, X_n)X_{n+1}). \tag{1.12}$$

Now, by projecting this equation onto $k[X_1, \ldots, X_{n+1}]/(1 - gX_{n+1})$, we get

$$g^\ell(X_1, \ldots, X_n) = \sum_{j=1}^{m} h_j(X_1, \ldots, X_{n+1}) g^\ell \times f_j(X_1, \ldots, X_n) \tag{1.13}$$

modulo $(1 - gX_{n+1})$, where $H_j = h_j(X_1, \ldots, X_{n+1})g^\ell \in k[X_1, \ldots, X_n]$, $\forall j$ if $\ell \gg 0$. Since $\{\overline{X}_1, \ldots, \overline{X}_n\}$ is algebraically free in $k[X_1, \ldots, X_{n+1}]/(1 - gX_{n+1})$, we conclude that, in fact, we have equality

$$g^\ell(X_1, \ldots, X_n) = \sum_{j=1}^{m} H_j(X_1, \ldots, X_n) \times f_j(X_1, \ldots, X_n) \tag{1.14}$$

in $k[X_1, \ldots, X_n]$. □

Remark 1.1.14. We could have used the homomorphism

$$k[X_1, \ldots, X_{n+1}] \to k(X_1, \ldots, X_n) : X_{n+1} \mapsto g,\ X_i \mapsto X_i,\ i \neq n+1 \tag{1.15}$$

at the end of the previous proof.

1.2 Lecture 3

1.2.1 Irreducible topological spaces

Example 1.2.1. The affine space k^m is irreducible, in fact if two open sets doesn't intersect, then there exist $f, g \in k[X_1, \ldots, X_m] \setminus \{0\}$ such that $D(f) \cap D(g) = 0 \Rightarrow fg = 0$, which is a contradiciton because $k[X_1, \ldots, X_m]$ is a domain.

1.2.2 Algebraic sets

Definition 1.2.2. We define an affine algebraic set $V \subset k^n$ to be a set of the form $V = V(S)$ for some $S \subset k[\underline{X}]$.

Notation 1.2.3. Let V be an affine algebraic set. We write

$$D_V(g) = \{p \in V : g(p) \neq 0\} = V \setminus V(g). \tag{1.16}$$

These open sets are called "principal open subsets".

Lemma 1.2.4. Any open subset $U \subset V$ of an affine algebraic set is quasicompact.

Proof: Consider a covering $U = \bigcup_{\lambda\in\Lambda} U_\lambda$. Observe that $V \setminus U = V(g_1, \ldots, g_m)$, so we have the equality $U = \bigcup_{i=1}^m D_V(g_i)$. In particular, we see that principal open subsets form a basis for the Zariski topology on V. Therefore, it's sufficient to prove that a principal open subset is quasicompact. So we assume $U = D_V(g)$. Furthermore, we have $U_\lambda = \bigcup_{i=1}^{n_\lambda} D_V(g_{\lambda,i})$ and

$$U = \bigcup_{\lambda\in\Lambda} \bigcup_{i=1}^{n_\lambda} D_V(g_{\lambda,i}). \tag{1.17}$$

So, without lost of generalizaiton, we can assume $U = \bigcup_{\lambda\in\Lambda} D_V(g_\lambda)$ from beginning. Now

$$\begin{aligned} D_V(g) = \bigcup_{\lambda\in\Lambda} D_V(g_\lambda) \iff \\ V \cap V(g) = V \cap V(\langle g_\lambda : \lambda \in \Lambda\rangle) \iff \\ V \cap V(g) = V \cap V(\langle g_\lambda : \lambda \in I\rangle)\ (I \subset \Lambda \text{ finite}). \end{aligned} \tag{1.18}$$

In the last equation we used Hilbert basis theorem: The sequence

$$\langle g_{\lambda_1}\rangle \subset \langle g_{\lambda_1}, g_{\lambda_2}\rangle \subset \cdots \tag{1.19}$$

must terminate at some point, because $k[\underline{X}]$ is Noetherian. Then we conclude that $D_V(g) = \bigcup_{\lambda\in I} D_V(g_\lambda)$. □

Corollary 1.2.5. From the proof of the previous lemma we see that every affine algebraic variety has a basis for its topology consisting of open affine subsets.

Lemma 1.2.6. Suppose that k is an infinity field, then open sets $U \subset k^n$ are dense in the Zariski topology.

Proof: Since basic open sets form a basis for the Zariski topology on k^n, it's sufficient to prove that $D(f)$ is dense in k^n. Let $\overline{D(f)} = V(g_1, \ldots, g_m)$, $m \geq 1$ and $x \in k^n$, then we have for all $i = 1, \ldots, m$:

$$f(x) \neq 0 \Rightarrow g_i(x) = 0 \Rightarrow fg_i(x) = 0,\ \forall x \in k^n. \tag{1.20}$$

Therefore, since k is infinity, we have $fg_i = 0 \in k[\underline{X}]$, $\forall i$ and $g_i = 0$, $\forall i$. Then we see that $\overline{D(f)} = k^n$. □

Definition 1.2.7. (Regular functions). Let $V \subset k^n$ be an affine algebraic set and $U \subset V$ be an open set. A map $\varphi : U \to k$ is said to be *regular* if for any $p \in U$, there exist $p \in U' \subset U$, polynomials $f, g \in k[\underline{X}]$ such that $g(q) \neq 0$, $\forall q \in U'$ and $\varphi|_{U'} = f/g|_{U'}$.

Notation 1.2.8. Let V be an affine algebraic set and let $U \subset V$ be an open set, we denote $\mathcal{O}_V(U) = \{U \overset{\text{regular}}{\to} k\}$.

Lemma 1.2.9. Let $V \subset k^n$ be an affine algebraic set and take $g \in k[X_1, \ldots, X_n] \setminus I(V)$, so $D_V(g) \neq \varnothing$. Then the set of regular functions $\varphi : D_V(g) \to k$ is of the form $f/g^m|_{D_V(g)}$ for some $f \in k[X_1, \ldots, X_n]$ and $m > 0$. Then we have an isomorphism

$$\widetilde{\psi} : k[V]_{\overline{g}} \to \mathcal{O}_V(D_V(g)) \tag{1.21}$$

where $\psi : k[V] \to \mathcal{O}_V(D_V(g))$ is the map that takes a polynomial and restricts it to $D_V(g)$ and $\widetilde{\psi}$ is the corresponding map induced by the universal property of localization.

Proof: First we have $D_V(g) = \bigcup_{i=1}^{\ell} D_V(g_i)$ and since φ is regular, there exists $f_i, h_i \in k[\underline{X}]$, $i = 1, \ldots, \ell$ such that

$$q \in D_V(g_i) \Rightarrow h_i(q) \neq 0 \text{ (so } q \in D_V(h_i)), \ \varphi\big|_{D_V(g_i)} = \frac{f_i}{h_i}\Big|_{D_V(g_i)}. \tag{1.22}$$

Now, we have two fundamental relations:

$$\begin{cases} D_V(g_i) \subset D_V(h_i), \ \forall i = 1, \ldots, \ell \\ D_V(g) = \displaystyle\bigcup_{i=1}^{\ell} D_V(g_i) \end{cases} \tag{1.23}$$

Using $V = V(F_1, \ldots, F_m)$ and Hilbert Nullstelensatz (1.1.13), we get

$$\begin{cases} \sqrt{(F_1, \ldots, F_m, g_i)} \subset \sqrt{(F_1, \ldots, F_m, h_i)}, \ \forall i = 1, \ldots, \ell \quad (*) \\ \sqrt{(F_1, \ldots, F_m, g)} = \sqrt{(F_1, \ldots, F_m, g_1, \ldots, g_\ell)} \quad (**) \end{cases} \tag{1.24}$$

From $(*)$, we deduce that $g_i^{n_i} = \sum_j A_j F_j + B_i h_i$ and restricting to V we get $g_i^{n_i} = B_i h_i$. Now, since $\varphi\big|_{D_V(g_i)} = \frac{B_i f_i}{g_i^{n_i}}$ and $D_V(g_i) = D_V(g_i^{n_i})$, we can change notation $B_i f_i \to f_i$ and $g_i^{n_i} \to g_i$.

Now we do a trick: since $g_j f_i - f_j g_i = 0$ on $D_V(g_i) \cap D_V(g_j) = D_V(g_i g_j)$, we get

$$g_i g_j (g_j f_i - f_j g_i) = 0 \text{ on } V. \tag{1.25}$$

Furthermore, we still have $\varphi|_{D_V(g_i)} = g_i f_i / g_i^2$. So we change notation again: $g_i^2 \to g_i$ and $g_i f_i \to f_i$, so we have (on V)

$$g_i f_j - g_j f_i = 0, \ \forall i, j = 1, \dots, \ell \quad (***). \tag{1.26}$$

Now, by equation $(**)$, there exist $m > 0$ such that

$$g^m = \sum_j A_j F_j + \sum_k B_k g_k \tag{1.27}$$

and we get on V:

$$g^m = \sum_k B_k g_k. \tag{1.28}$$

Now, by using equation $(***)$, we have (on V)

$$f_i g^m = \sum_k B_k f_i g_k = \sum_k B_k f_k g_i = (\sum_k B_k f_k) g_i. \tag{1.29}$$

So if $q \in D_V(g_i)$, we have $\varphi(q) = \frac{f_i}{g_i}(q) = \frac{f}{g^m}(q)$, where we set $f = \sum_k B_k f_k$. This proves that $\widetilde{\psi}$ is surjective.

To prove injectivity, we observe that $f/g^m|_{D_V(g)} = 0 \iff f|_{D_V(g)} = 0 \iff fg|_V = 0$, which is equivalent to $fg \in I(V)$. Therefore $\overline{g} \cdot \overline{f} = 0$, which implies $\overline{f}/\overline{g}^m = 0 \in A_{\overline{g}}$ by definition. □

Corollary 1.2.10. Since $V = D_V(1)$, we have

$$\mathcal{O}_V(V) \cong k[V]_{(1)} \cong k[V]. \tag{1.30}$$

Remark 1.2.11. Normally, it's not obvious how to describe the set of units $\mathcal{O}_V(D_V(G))^*$ even if V is a irreducible affine algebraic set. For example, if $V = k^1$, then $k[V] = k[X]$. So, for $G = X(X-1)$ we have $\mathcal{O}_V(D_V(G))^* \cong k^* \langle X^m, (X-1)^m : m \in \mathbb{Z} \rangle$, by the identification

$$\mathcal{O}_V(D_V(G)) \cong k[X]_{X(X-1)} \cong k[X, X^{-1}, (X-1)^{-1}]. \tag{1.31}$$

Definition 1.2.12. A morphism between two affine algebraic sets $V_1 \subset k^n$ and $V_2 \subset k^m$ is a function $\varphi : V_1 \to V_2$ such that for any open set $U \subset V_2$, we have

$$\varphi_U^{\#} : \mathcal{F}(U, k) \xrightarrow{-\circ\varphi} \mathcal{F}(\varphi^{-1}(U), k), \ \varphi_U^{\#}(\mathcal{O}_{V_1}(U)) \subset \mathcal{O}_{V_2}(\varphi^{-1}(U)). \tag{1.32}$$

Using more fancing words, the previous definition says that a continuous map $\varphi : V_1 \to V_2$ is a morphism if the composition map $\varphi_U^{\#} : \mathcal{F}(U,k) \overset{-\circ\varphi}{\to} \mathcal{F}(\varphi^{-1}(U),k)$ induces a sheaf morphism

$$\varphi^{\#}(U) : \mathcal{O}_{V_2} \to f_*\mathcal{O}_{V_1}. \tag{1.33}$$

We use the notation $f_*\mathcal{O}_{V_1}$ for direct image sheaf, it's defined by $f_*\mathcal{O}_{V_1}(U) = \mathcal{O}_{V_1}(f^{-1}(U))$.

1.3 Lecture 4

Definition 1.3.1. A function space is a topological space X with a sheaf $\mathcal{O}_X$ such that for any open set $U \subset X$, we have

(1) $\mathcal{O}_X(U) \overset{\text{subring}}{\subset} \{U \overset{\text{continuous}}{\to} k\}$

(2) Let $f \in \mathcal{O}_X(U)$, then f has an inverse in $\mathcal{O}_X(U)$ if and only if $f(q) \neq 0,\ \forall q \in U$.

Lemma 1.3.2. An affine algebraic set $V \subset k^n$ equiped if the sheaf of regular functions is a function space.

Proof: Let $U \subset X$ be an open set and $f \in \mathcal{O}_X(U)$. Then, by definition, there exist an open cover $U = \bigcup_\lambda U_\lambda$ such that $f|_{U_\lambda} = f_\lambda/g_\lambda$, where $g_\lambda(q) \neq 0\ \forall q \in U_\lambda$. To prove that f is continuous, it's sufficient to prove that $f^{-1}(a) \subset V$ is closed for all $a \in k$. So we prove that $(f|_{U_\lambda})^{-1}(a)$ is closed in V for all λ, which can be verified by the equality

$$V \cap V(f_\lambda - ag_\lambda) = (f|_{U_\lambda})^{-1}(a). \tag{1.34}$$

The second item is clear. □

Definition 1.3.3. Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be two function spaces. We define a morphism $X \to Y$ to be a a continuous map $f : X \to Y$ which induces a sheaf map $f^{\#} : \mathcal{O}_Y \to f_*\mathcal{O}_Y$ in the sence that if $f^{\#}(U)(g) = g \circ f$ for $g \in \mathcal{O}_Y(U)$, then $f^{\#}(U)(\mathcal{O}_Y(U)) \subset \mathcal{O}_X(f^{-1}(U))$.

Remark 1.3.4. Being a morphism of function spaces is a local property. In fact, let $f : X \to Y$ be a morphism of function spaces and suppose that $Y = \bigcup_\lambda V_\lambda$ and for each λ:

$$f|_{f^{-1}(V_\lambda)} : f^{-1}(V_\lambda) \to V_\lambda \tag{1.35}$$

is a morphism. Then f is a morphism.

Definition 1.3.5. An affine algebraic variety is a function space $(X, \mathcal{O}_X)$ isomorphic to an affine algebraic set with Zariski topology and sheaf of regular functions.

Definition 1.3.6. Let $(X, \mathcal{O}_X)$ be a function space, and $U \subset X$ be an open set with induced topology and $\mathcal{O}_U = \mathcal{O}_X|_U$. Then $(U, \mathcal{O}_U)$ is a function space called induced open subspace.

Definition 1.3.7. An abstract algebraic variety is a function space $(X, \mathcal{O}_X)$ such that there exist an open covering $X = \bigcup_\lambda U_\lambda$ with the property that for each λ, the induced open subspace $(U_\lambda, \mathcal{O}_{U_\lambda})$ is an affine algebraic variety.

Remark 1.3.8. In many texts we have the additional hypothesis of irredutibility in the definition of algebraic varieties.

Definition 1.3.9. An affine algebraic variety is said to be quasi-affine if it's isomorphic, as a function space, to an induced open subspace of an affine algebraic variety.

Lemma 1.3.10. Let $(X, \mathcal{O}_X)$ be a function space and $Y \subset k^n$ be a quasi-affine algebraic set. Then $f : X \to Y$ is a morphism if and only $f = (g_1, \ldots, g_n)$ with $g_i \in \mathcal{O}_X(X)$. Here, we think Y as an induced open subspace of an affine algebraic set.

Proof: ($\Rightarrow$) The coordinate functions $x_i|_Y : Y \to k$ are regular functions, by definition. Now $g_i = x_i \circ f = f^\#(Y)(x_i) \in \mathcal{O}_X(f^{-1}(Y)) = \mathcal{O}_X(X)$, because $x_i \in \mathcal{O}_Y(Y) = \{Y \overset{\text{regular}}{\to} k\}$.

($\Leftarrow$) A closed set of Y is of the form $V(F) \cap Y$ (would not be of the form $V(F_1, \ldots, F_m)$ in general ?). However $f^{-1}(V(F) \cap Y) = \{p \in X : F(g_1, \ldots, g_n)(p) = 0\}$ which is closed in X provided we show that $F(g_1, \ldots, g_n)$ is continuous. However, since $\mathcal{O}_X(X)$ is a ring, we have $F(g_1, \ldots, g_n) \in \mathcal{O}_X(X)$. Furthermore, if $U \subset Y$ is an open set and $\varphi : U \to k$ is regular, we need to show that $f^\#(U)(\varphi) \in \mathcal{O}_X(f^{-1}(U))$. Let $U' \subset U$ be such that $\varphi|_{U'} = F/G$, $G(q) \neq 0 \; \forall q \in U'$. Then we have

$$f^\#(U')(\varphi) = \varphi \circ f|_{f^{-1}(U')} = \left.\frac{F(g_1, \ldots, g_n)}{G(g_1, \ldots, g_n)}\right|_{f^{-1}(U')}. \tag{1.36}$$

Furthermore

$$G \circ f(q) = G(g_1, \ldots, g_n)(q) \neq 0, \; \forall q \in f^{-1}(U'). \tag{1.37}$$

Since $G(g_1,\ldots,g_n)|_{f^{-1}(U')} \in \mathcal{O}_X(f^{-1}(U'))$, the previous equation shows that the inverse $1/G(g_1,\ldots,g_n)$ also is an element of $\mathcal{O}_X(f^{-1}(U'))$. Now, let $U = \bigcup_\lambda U_\lambda$, $\varphi|_{U_\lambda} = F_\lambda/G_\lambda$, then we see that

$$f^{\#}(U_\lambda)(\varphi|_{U_\lambda}) \subset \mathcal{O}_X(f^{-1}(U_\lambda)). \tag{1.38}$$

The glueing axiom shows that $f^{\#}(U)(\varphi) \in \mathcal{O}_X(f^{-1}(U))$. □

Lemma 1.3.11. Let $V \subset k^n$ be an affine algebraic set and $D_V(F) \subset V$ an affine algebraic set. Then the induced open subspace $(D_V(F), \mathcal{O}_{D_V(F)})$ is a quasi-affine algebraic variety, by definition. However, we have more, it's also an affine algebraic variety.

Proof: Let $V = V(G_1,\ldots,G_m)$ and define the map

$$\varphi : D_V(F) \to V(x_0F - 1, G_1,\ldots,G_m) \subset k^{n+1} : (a_1,\ldots,a_n) \mapsto (\frac{1}{F(a_1,\ldots,a_n)}, a_1,\ldots,a_n) \tag{1.39}$$

By the lemma (1.3.10), we see that φ is a morphism. Furthermore, it has a set theoretical inverse given by the projection

$$\psi : V(x_0F - 1, G_1,\ldots,G_m) \to D_V(F) : (a_0,\ldots,a_n) \mapsto (a_1,\ldots,a_n). \tag{1.40}$$

By the same lemma, this map is also a morphism. Since $f\psi = \mathbb{1}_{D_V(x0F-1,\ldots)}$ and $\psi f = \mathbb{1}_{D_V(F)}$, we see that φ is an isomorphism (the induced sheaf morphism would be an isomorphism by construction). □

Remark 1.3.12. Let $(X, \mathcal{O}_X)$ be an affine algebraic variety and $\varphi : (V, \mathcal{O}_V) \to (X, \mathcal{O}_X)$ an isomorphism, where $V \subset k^n$ is an algebraic set. Then this isomorphism, induces another isomorphism

$$(D_V(F), \mathcal{O}_{D_V(F)}) \to (X_f, \mathcal{O}_{X_f}) \tag{1.41}$$

where $F = f \circ \varphi$ and $X_f = \varphi(D_V(F)) = \{p \in X : f(p) \neq 0\}$.

Corollary 1.3.13. A quasi-affine algebraic variety is an abstract algebraic variety.

Proof: A quasi-affine algebraic variety has a topological basis consisting of principal open subsets, which are affine algebraic varieties.

1.4 Lecture 5 and 6

Theorem 1.4.1. Suppose that we have the data:

- $(X_\lambda)_{\lambda\in\Lambda}$ function spaces over k
- $\forall\lambda,\mu\in\Lambda$ let $U_{\lambda\mu}\subset X_\lambda$
- $\forall\lambda,\mu\in\Lambda$ an isomorphism
$$\varphi_{\lambda\mu}: U_{\lambda\mu}\to U_{\mu\lambda} \tag{1.42}$$

Furthermore, assume that

- $U_{\lambda\lambda}=X_\lambda,\ \varphi_{\lambda\lambda}=\mathbb{1}_{X_\lambda}$
- $\forall\lambda,\mu,\nu\in\Lambda$, we have
$$\begin{aligned}&\varphi_{\lambda\mu}(U_{\lambda\mu}\cap U_{\lambda\nu})\subset U_{\mu\lambda}\cap U_{\mu\nu}\\&\varphi_{\mu\nu}\varphi_{\lambda\mu}|_{U_{\lambda\mu}\cap U_{\lambda\nu}}=\varphi_{\lambda\nu}|_{U_{\lambda\mu}\cap U_{\lambda\nu}}\end{aligned} \tag{1.43}$$

Then, $X=\coprod_{\lambda\in\Lambda}X_\lambda/\sim$, where we put $(\lambda,x)\sim(\mu,x')$ if and only if $\varphi_{\lambda\mu}(x)=x'$. One can verify that this defines an equivalence relation. Furthermore, X has a topology defined as the small topology such that all the maps $i_\lambda: X_\lambda\to X: x\mapsto[(\lambda,x)]$ are continuous. We define a sheaf

$$\mathcal{O}_X(U)=\{f: U\to k: fi_\lambda\in\mathcal{O}_{X_\lambda}(i_\lambda^{-1}(U)),\ \forall\lambda\in\Lambda\}. \tag{1.44}$$

With this sheaf, $(X,\mathcal{O}_X)$ is a function space satisfying the following universal property: for any function space Y and morphisms $j_\lambda: X_\lambda\to Y$such that:

$$j_\mu\varphi_{\lambda\mu}=j_\lambda|_{U_{\lambda\mu}}, \tag{1.45}$$

there exist an unique morphism $f: X\to Y$ such that $fi_\lambda=j_\lambda$ (morphisms here, means morphisms of function spaces !). Furtherore, since $i_\mu^{-1}i_\lambda(X_\lambda)=U_{\mu\lambda}\subset X_\mu,\ \forall\mu$, then $i_\lambda(U_\lambda)$ is open and i_λ are isomorphisms onto its images. So, if each X_λ is an affine algebrai variety, then X is an abstract variety.

Proof: Tedious. □

Definition 1.4.2. An abstract variety X is separable if for any pair of affine open sets $U, V \subset X$, we have

(1) $U \cap V$ is affine

(2) The induced map

$$\mathcal{O}(U) \otimes \mathcal{O}(V) \to \mathcal{O}(U \cap V) : f \otimes g \mapsto f|_{U\cap V} \cdot g|_{U\cap V} \tag{1.46}$$

is surjective.

Example 1.4.3. Let $X_0 = X_1 = k^1$ and $U_{01} = U_{10} = k^1 - \{0\}$. Define a map $\varphi_{01} : U_{01} \to U_{10} : x \mapsto 1/x$. We claim that $X = \mathbb{P}^1$ in the variety obtained by glueing X_0 and X_1 through φ_{01}. In fact, for $i = 0, 1$, let $g_i : X_i \to Y$ be two morphisms such that

$$g_1\varphi_{01} = g_0\big|_{U_{01}}. \tag{1.47}$$

That is, $g_1(1/x) = g_0(x)$. Then the following map is well defined:

$$\begin{aligned} g :& X \to Y \\ & [1 : y] \mapsto g_0(y) \\ & [x : 1] \mapsto g_1(x) \end{aligned} \tag{1.48}$$

Moreover, by construction we have $g|_{U_i} = g_i\varphi_i$, where $\varphi_i : U_i \to k^1$ (here, $U_i = \{[x_0, x_1] : x_i \neq 0\}$) are the standard charts of the projective plane X. So we se that g is locally a morphism, i.e., it's a morphism.

1.5 Lecture 6,7,8

Remark 1.5.1. Take the product topology on $k^1 \times k^1$, where k^1 is equiped with Zariski topology. Let $U = k^1 - S$ and $V = k^1 - T$ (S, T finite) be two open sets. Then $U \times V = k^1 \times k^1 - (S \times k^1 \cup k^1 \times T)$ and in general, for a familiy of open sets U_λ, V_λ, we have the equality.

$$\begin{aligned} (\bigcup_\lambda U_\lambda \times V_\lambda)^c &= \bigcap_\lambda S_\lambda \times k^1 \cup k^1 \times T_\lambda \\ &= (\bigcap_\lambda S_\lambda \times k^1) \cup (k^1 \times \bigcap_\lambda T_\lambda) \end{aligned} \tag{1.49}$$

where S_λ, T_λ are all finite. So any closed set $F \subset k^1 \times k^1$ is of the form

$$F = S \times k^1 \cup k^1 \times T,\ \#S, \#T < \infty \tag{1.50}$$

in the product topology. In particular, if $\#k = \infty$, then $V(X, Y) = \{0\}$ is not closed in the product topology.

TODO: remains to study lect 8, Nullstelensatz for projective spaces and "universal compactness" theorem.

1.5.1 Projective varieties

We define $\mathbb{P}^m = (k^{m+1}\backslash\{0\})/\sim$. As usual, define charts $\varphi_i : D_+(X_i) \to k^m : [x_0, \ldots, x_m] \mapsto (\frac{x_0}{x_i}, \ldots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \ldots, \frac{x_m}{x_i})$. Then we see that the transition maps $\varphi_i\varphi_j^{-1} : D(X_i) \subset k^m \to D(X_j) \subset k^m$ are isomorphisms of affine varieties. Furthermore, $\mathbb{P}^m$ has a topology induced by the open affine covering $\mathbb{P}^m = \bigcup_{i=0}^m D_+(X_i)$ where we set $U \subset \mathbb{P}^m$ to be open if $U \cap D_+(X_i) \subset D_+(X_i)$ is open $\forall i$. With this definition it's clear that $D_+(X_i)$ is open, because $\varphi_i(D_+(X_iX_j)) = D(X_j) \subset k^m$ is open $\forall j$. The sheaf of regular function is defined by using the charts:

$$\mathcal{O}_{\mathbb{P}^m}(U) = \{f : U \to k : f\varphi_i^{-1} \in \mathcal{O}_{k^m}(\varphi_i(U)),\ \forall i\} \tag{1.51}$$

In this condition, we see that $\mathbb{P}^m$ is an abstract variety.

Proposition 1.5.2. Let $\varphi_{ij} = \varphi_j\varphi_i^{-1} : \varphi_i(D(X_iX_j)) \to \varphi_j(D(X_iX_j))$. Write $U_{ij} = \varphi_i(D(X_iX_j)) \subset k^m$. Then $\varphi_{ij} : U_{ij} \to U_{ji}$ satisfies all the conditions of theorem (1.4.1). We claim that $\mathbb{P}^m$ is isomorphic to the variety obtatined by glueing all k^m through the isomorphisms φ_{ij}.

Proof: We have a well defined bijection

$$\mathbb{P}^m \to X = \coprod_{i=0}^m \{i\} \times k^m \Big/ \sim : D_+(X_i) \ni [x_0, \ldots, x_m] \mapsto [\varphi_i([x_0, \ldots, x_m])_i] \tag{1.52}$$

where we write $(\underline{x})_i$ for an element of $\{i\} \times k^m$. It remains to show that this map is an isomorphism. However, at the level of charts we have the identity map. □

Proposition 1.5.3. The projective space $\mathbb{P}^m$ is irreducible.

Proof: Let $P, Q \in \mathbb{P}^m$, then there exist $H \in (k^{m+1})^*$ such that $H(P), H(Q) \neq 0$, so $P, Q \in D_+(H)$. Now, let $U, V \subset \mathbb{P}^m$ be two open sets, points $P \in U, Q \in V$ and H be such that $P, Q \in D_+(H)$. Then $U \cap D_+(H), V \cap D_+(H)$ are both nonempty open sets in the irreducible affine space $D_+(H) \cong k^m$, so they must intersetc. □

Remark 1.5.4. We know that $\mathbb{P}^1 - \{(0:1)\}$ is affine. We will show that $U = \mathbb{P}^2 - \{(0:0:1)\}$ is not affine. Let $f \in \mathcal{O}_U(U)$, then by definition of regular function we have

$$f|_{U_0}(1 : x_1 : x_2) = F_0(x_1, x_2),\ f|_{U_1}(x_0, 1, x_2) = F_1(x_0, x_2). \tag{1.53}$$

On the intersection $U_0 \cap U_1$, we have

$$F_0(x_1, x_2) = f(1 : x_1 : x_2) = f(x_1^{-1} : 1 : x_1x_2) = F_1(x_1^{-1}, x_1^{-1}x_2) \in k[X_1, X_2]. \tag{1.54}$$

So f must be a constant and $\mathcal{O}_U(U) \cong k$.

1.5.2 Projective Nullstelensatz

Proposition 1.5.5. Let $I \subset k[X_0, \ldots, X_m]$ be a ideal generated by homogeneous polynomial. Then $F = \sum_i F_i \in I \iff F_i \in I \ \forall i$ where F_i are the homogeneous components of F. In each case we say that the ideal I is *homogeneous*.

Proposition 1.5.6. An ideal I is homogeneous if and only if $\sqrt{I}$ is homogeneous.

Notation 1.5.7. Let $I \subset k[X_0, \ldots, X_m]$ be a homogeneous ideal, then we denote

$$V_p(I) = \{[x_0, \ldots, x_m] \in \mathbb{P}^m : F(x_0, \ldots, x_m) = 0, \ \forall F \in I \text{ homogenous}\}. \tag{1.55}$$

Furthermore, if $V \subset \mathbb{P}^m$ is a subset, we write

$$I_p(V) = \langle S \rangle, \ S = \{F \in k[X_0, \ldots, X_m] : F \text{ is homogeneous and } F(p) = 0, \ \forall p \in V\}. \tag{1.56}$$

Theorem 1.5.8. Let $I \subset k[X_0, \ldots, X_m]$ be a homogeneous ideal. Then

(1) $V_p(I) = \varnothing \iff \sqrt{I} \supset (X_0, \ldots, X_m)$

(2) If $V_p(I) \neq \varnothing$, then $I_p V_p(I) = \sqrt{I}$

Proof: For the first statement, we observe that $V_p(I) = \varnothing \iff V_a(I) = \{0\}$. However, by the Nullstelensatz we have

$$\sqrt{I} = I_a V_a(I) = I_a(\{0\}) = (X_0, \ldots, X_m). \tag{1.57}$$

For the second statement, we observe that if $V_p(I) \neq \varnothing$, then

$$\begin{aligned} I_p V_p(I) &= \langle F \text{ homogeneous } : F[x_0, \ldots, x_m] = 0, \ \forall [x_0, \ldots, x_m] \in V_p(I) \rangle \\ &= \langle F \text{ homogeneous } : F[x_0, \ldots, x_m] = 0, \ \forall (x_0, \ldots, x_m) \in V_a(I) \setminus \{0\} \rangle \\ &= \langle F \text{ homogeneous } : F(x_0, \ldots, x_m) = 0, \ \forall (x_0, \ldots, x_m) \in V_a(I) \rangle \\ &= \langle F \text{ homogeneous } : F \in I_a V_a(I) \rangle \\ &= \langle F \text{ homogeneous } : F \in \sqrt{I} \rangle = \sqrt{I}. \end{aligned} \tag{1.58}$$

The last line follows because $\sqrt{I}$ is a homogenous ideal. □

1.5.3 Segre embedding

Remark 1.5.9. Here, to ease notation, we will write $x = [x_0, \ldots, x_m]$ sometimes.

The Segre embedding is the map

$$\begin{aligned} s : \mathbb{P}^m \times \mathbb{P}^n &\to \mathbb{P}^{(m+1)(n+1)-1} \\ ([x_0, \ldots, x_m], [y_0, \ldots, y_n]) &\mapsto \begin{bmatrix} x_0y_0 & x_0y_1 & \ldots & x_0y_n \\ x_1y_0 & x_1y_1 & \ldots & x_1y_n \\ \vdots & \vdots & \vdots & \vdots \\ x_my_0 & x_my_1 & \ldots & x_my_n \end{bmatrix} \end{aligned} \tag{1.59}$$

where we parametrize $\mathbb{P}^{(m+1)(n+1)-1}$ with projective coordinates $z_{ij},\ 0 \leq i \leq m, 0 \leq j \leq n$. We claim that $s(\mathbb{P}^m \times \mathbb{P}^n)$ is a closed subset. In fact, if we let $z_{ij} = x_iy_j$, then we have $z_{ij}z_{pq} = z_{iq}z_{pj}$ and

$$s(\mathbb{P}^m \times \mathbb{P}^n) \subset V = V_p(z_{ij}z_{pq} - z_{pj}z_{iq},\ \forall p, q, i, j) \quad (*). \tag{1.60}$$

Conversely, if we take $z \in V$, then $z_{ij} \neq 0$ for some i, j. So we assume $z_{ij} = 1$. Then we see that $z_{pq} = z_{pj}z_{iq}$, so we can take $x_p = z_{pj}$ and $y_q = z_{iq}$ to conclude that $s(x, y) = z$. Then we conclude the equality of $(*)$. Furthermore, we claim that this map is injective. Let $s(x, y) = s(x', y')$, then $x_py_q = \lambda x'_py'_q$. Fix q such that $y_q \neq 0$, then $x_p = (\frac{\lambda y'_q}{y_q})x'_p,\ \forall p$ and $x = x'$. In a similar way we see that $y = y'$.

As a corollary, we see that s mapst $\mathbb{P}^m \times \mathbb{P}^n$ bijectively to a closed subset $V \subset \mathbb{P}^{(m+1)(n+1)-1}$. We put a structure of abstract variety on $\mathbb{P}^m \times \mathbb{P}^n$ in such a way that s is an isomorphism onto V. So we see for example, that

$$\mathcal{O}_{\mathbb{P}^m \times \mathbb{P}^n}(U) = (- \circ s)(\mathcal{O}_V(s(U))). \tag{1.61}$$

In a concrete way, a section $s \in \mathcal{O}_{\mathbb{P}^m \times \mathbb{P}^n}$ is a quotient of bihomogeneous polynomials of bi degree (d, d) each one, for some $d \geq 0$. In fact, if $F(z) \in k[\underline{z}]$ is homogeneous, then

$$F(z) = \sum a_\alpha \prod_{i,j} z_{ij}^{\alpha_{ij}},\ (i, j) \in \{0, \ldots, m\} \times \{0, \ldots, n\}. \tag{1.62}$$

Therefore

$$\begin{aligned} F \circ s(x, y) &= \sum a_\alpha \prod_{i,j} x_i^{\alpha_{ij}} y_j^{\alpha_{ij}} \\ &= \sum a_\alpha (\prod_i x_i^{\sum_j \alpha_{ij}})(\prod_j y_j^{\sum_i \alpha_{ij}}) \end{aligned} \tag{1.63}$$

where $\sum_i \sum_j \alpha_{ij} = \sum_j \sum_i \alpha_{ij} = |\alpha| = d$, the degree of F. It's easy to see that a quotient of bihomogeneous polynomials of bidegree (d, e) each one, is also a section of $\mathcal{O}_{\mathbb{P}^m \times \mathbb{P}^n}$. In fact, let G/H be a quotient of two bihomogeneous polynomials of bidegree (d, e) each one. Then, in $D(x_i) \times D(y_j)$ we have

$$\frac{G}{H} = \frac{Gx_i^e y_j^d}{Hx_i^e y_j^d}. \tag{1.64}$$

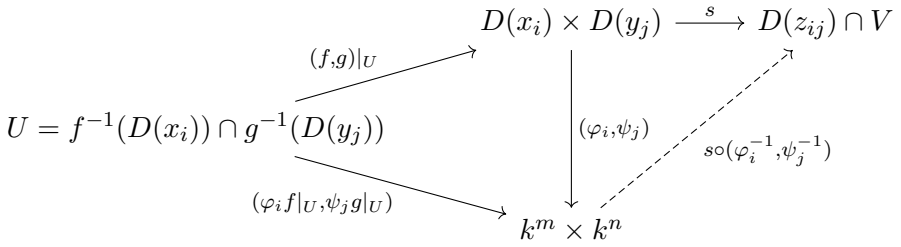
So we can assume both G, H to be of the same bidegree $(d+e, d+e)$. Moreover, let F be a bihomogeneous polynomial of bidegree (d, d), then

$$\begin{aligned} F(x,y) &= \sum a_{\alpha\beta} x^\alpha y^\beta \\ &= \sum a_{\alpha\beta} x_{i_1} y_{j_1} \cdots x_{i_d} y_{j_d} \ (1 \le i_k \le m, 1 \le j_k \le n) \\ &= \sum a_{\alpha\beta} z_{i_1 j_1} \cdots z_{i_d j_d} \end{aligned} \tag{1.65}$$

is homogeneous of degree d.

Lemma 1.5.10. The variety $\mathbb{P}^m \times \mathbb{P}^n$ constructed via the Segre embedding is a product.

Proof: Let $f : Z \to \mathbb{P}^m$ and $g : Z \to \mathbb{P}^n$ be morphisms. We draw the following diagram



We see that $\varphi_i f|_U, \psi_j g|_U$ are morphisms, so $(\varphi_i f|_U, \psi_j g|_U)$ is a morphism because it's a mapping into a product of affine varieties and we can apply the affine criteria. So, if we show that $s \circ (\varphi_i^{-1}, \psi_j^{-1})$ is a morphism, we are done. However, we see that

$$\xi_{ij} \circ s \circ (\varphi_i^{-1}, \psi_j^{-1}) : k^m \times k^n \to k^{(m+1)(n+1)} : (x,y) \mapsto (\dots, x_i, \dots, y_j, \dots, x_i y_j, \dots) \tag{1.66}$$

which is clearly a morphism. □

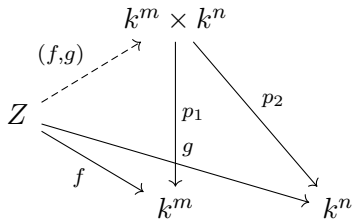
Remark 1.5.11. The above proof also shows that $(\varphi_i, \psi_j) : D(x_i) \times D(y_j) \to k^m \times k^n$ is a morphism. Furthermore, let $X \subset \mathbb{P}^m, Y \subset \mathbb{P}^n$ be two closed subsets. Then, is $X \cap D(x_i) \times Y \cap D(y_j) \subset D(x_i) \times D(y_j)$ closed ?. We have that $\varphi_i(X \cap D(x_i)) \subset k^m, \psi_j(Y \cap D(y_j)) \subset k^n$ are closed. Therefore, the product

$$\varphi_i(X \cap D(x_i)) \times \psi_j(Y \cap D(y_j)) = (\varphi_i, \psi_j)(X \cap D(x_i) \times Y \cap D(y_j)) \tag{1.67}$$

is closed in $k^m \times k^n$. Here we use the standard observation that $V(S) \times V(T) = V(S \cup T)$, where $S \subset k[\underline{X}], T \subset k[\underline{y}]$ with both rings sitting inside $k[\underline{x}, \underline{y}]$. Since (φ_i, ψ_j) is a homeomorphism, this proves that the answer is positive.

Corollary 1.5.12. Let $X \subset \mathbb{P}^m, Y \subset \mathbb{P}^n$ be two closed subsets. Then $X \times Y \subset \mathbb{P}^m \times \mathbb{P}^n$ is closed.

Remark 1.5.13. The product of the affine spaces k^m and k^n if $k^m \times k^n \cong k^{m+n}$. So, if $f : Z \to k^m, g : Z \to k^n$ are morphisms, then we draw the diagram.



However, f being a morphism means that $x_i \circ f$ and $y_j \circ g$ are morphims for all i, j ($1 \leq i \leq m, 1 \leq j \leq n$), where x_i and y_j are the standard coordinate functions. However, the set of coordinate functions of the product $k^m \times k^n$ is $(x_1, \ldots, x_m, y_1, \ldots, y_n)$ because of the identification $k^m \times k^n \cong k^{m+n}$. So, again we see that

$$(x_1 \circ f, \ldots, x_m \circ f, y_1 \circ g, \ldots, y_n \circ g) = (f, g) \tag{1.68}$$

is a morphism. Here we used lemma (1.3.10) several times.

1.5.4 Grassmanians and Flag varieties

Let $\mathrm{Gr}_d(V)$ be the set of $d-$dimensional subspaces of V. We have the following embedding

$$\psi : \mathrm{Gr}_d(V) \hookrightarrow \mathbb{P}(\bigwedge^d V) : W = \mathrm{Span}(w_1, \ldots, w_d) \mapsto [w_1 \wedge \cdots \wedge w_d]. \tag{1.69}$$

The map is well defined, in fact, if $\{w_i\}_{i=1}^d$ and $\{w_i'\}_{i=1}^d$ are two basis of W and $w_i' = \sum_j b_{ij} w_i$, then $w_1' \wedge \cdots \wedge w_d' = \det(b_{ij})_{i,j=1}^d w_1 \wedge \cdots \wedge w_d$. Now, let $v_1, \ldots, v_n$ be a basis of V and $u_1, \ldots, u_d \in V$ be arbitrary vectors. Write $u_i = \sum_j a_{ij} v_j$, then

$$\begin{aligned} u_1 \wedge \cdots \wedge u_d &= (\sum_{j_1=1}^n a_{1j_1} v_{j_1}) \wedge \cdots \wedge (\sum_{j_d=1}^n a_{dj_d} v_{j_d}) \\ &= \sum_{j_1=1}^n \cdots \sum_{j_d=1}^n a_{1j_1} \cdots a_{dj_d} v_{j_1} \wedge \cdots \wedge v_{j_d} \\ &= \sum_{j_1 < \cdots < j_d} \det(a_{ij_k})_{i,k=1}^d v_{j_1} \wedge \cdots \wedge v_{j_d} \end{aligned} \tag{1.70}$$

So we see that $u_1 \wedge \cdots \wedge u_d = 0 \iff \mathrm{Rank}(a_{ij}) < d \iff \{u_1, \ldots, u_d\}$ is linearly dependent. So, if

$$w_1 \wedge \cdots \wedge w_d = \lambda w_1' \wedge \cdots \wedge w_d' \neq 0, \tag{1.71}$$

then $u \wedge w_1 \wedge \cdots \wedge w_d = 0 \iff u \wedge w_1' \wedge \cdots \wedge w_d' = 0$, i.e., $u \in \mathrm{Span}\{w_i\}_{i=1}^d \iff u \in \mathrm{Span}\{w_i'\}_{i=1}^d$. Now, let $\mathcal{J} = \{(j_1, \ldots, j_d) : 1 \leq j_1 < \cdots < j_d \leq n\}$ and if $J = (j_1, \ldots, j_d) \in$

$\mathcal{J}$, write $v_J = v_{j_1} \wedge \cdots \wedge v_{j_d}$. Then we have the basis $\{v_J\}_{J\in\mathcal{J}}$ of $\bigwedge^d V$ and we write X_J for the corresponding coordinate functions.

Let $W \in \mathrm{Gr}_d(V)$, we see that $\psi(W) \in D(X_{(1,\dots,d)})$ if and only if $\det(a_{ij})_{i,j=1}^d \neq 0$. In this case, by row reduction, we can assume that $a_{ij} = \delta_{ij},\ (1 \leq i,j \leq d)$, so W has a basis consisting of vectors of the form $w_i = v_i + \sum_{j>d} a_{ij}v_j,\ i = 1,\dots,d$. Now define J_{pq} as the tuple $(j_1,\dots,j_d)$ such that $j_k = k\ (k < p),\ j_k = k+1,\ (p \leq k < d)$ and $j_d = q \in \{d+1,\dots,n\}$. With this notation, we see that

$$a_{pq} = \pm \det(a_{ij_k})_{j,k=1}^d = \pm X_{J_{pq}}(w_1 \wedge \cdots \wedge w_d). \tag{1.72}$$

So, for any $J = (j_1,\dots,j_d) \in \mathcal{J}$, the expression $\det(a_{ij_k})_{i,k=1}^d$ is a polynomial $f_J(X) \in k[X_{J_{pq}} : 1 \leq p \leq d, d < q \leq n] \subset k[X_J : J \in \mathcal{J} \setminus \{(1,\dots,d)\}]$ and we see that

$$\begin{aligned}&\varphi_{(1,\dots,d)}(\mathrm{Im}(\psi) \cap D(X_{(1,\dots,d)}))\\ &\quad = V(X_J - f_J(X) : \forall J = (j_1,\dots,j_d) \in \mathcal{J} \setminus \{(1,\dots,d), J_{pq} : p,q\})\end{aligned} \tag{1.73}$$

which is closed, so $\mathrm{Im}(\psi) \cap D(X_{(1,\dots,d)})$ is closed. With the same argument applied to the other principal open subsets $D(X_J)$, we see that $\mathrm{Im}(\psi)$ is closed. It's important to no note that the row echelon form of $A = (a_{ij})$ gives an affine coordinate expression for $\psi(W) \in U_{(1,\dots,d)}$:

$$\varphi_{(1,\dots,d)}\psi(W) = (f_J(a))_{J\in\mathcal{J}\setminus\{(1,\dots,d)\}} \tag{1.74}$$

where $f_J(a) = \det(a_{ij_k})_{i,k=1}^d,\ J = (j_1,\dots,j_d)$.

Now we define the Flag variety $\mathcal{F}(V)$, it's defined as the n-tuple of subspaces

$$(F_1,\dots,F_n) \in \mathrm{Gr}_1(V) \times \cdots \times \mathrm{Gr}_n(V) \tag{1.75}$$

such that $F_i \subset F_{i+1},\ 1 \leq i < n$. It's a closed subset, so it's a projective variety in the above product. We prove this in two steps. The first one is to prove that

$$\{(F,F') : F \subset F'\} \subset \mathrm{Gr}_d(V) \times \mathrm{Gr}_{d+1}(V) \tag{1.76}$$

is a closed subset. As before, we simplify the problem by showing that the intersection

$$\{(F,F') : F \subset F'\} \cap U_{(1,\dots,d)} \times U_{(1,\dots,d,d+1)} \quad (*) \tag{1.77}$$

is closed. So let F and F' be represented by matrices $(a_{ij}) \in \mathrm{Mat}_{d,n}(k)$ and $(b_{pq}) \in \mathrm{Mat}_{d+1,n}(k)$ in row echelon form, i.e., $a_{ij} = \delta_{ij},\ j \leq d$ and $b_{pq} = \delta_{pq},\ q \leq d+1$. In this condition $F \subset F'$ if and only if the matrix

$$M = \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \cdots & \vdots \\ b_{d+1,1} & \cdots & b_{d+1,n} \\ a_{11} & \cdots & a_{1n} \\ \vdots & \cdots & \vdots \\ a_{d1} & \cdots & a_{dn} \end{pmatrix} \tag{1.78}$$

has rank $d+1$. In particular, this means that any $d+2$ minor of M must vanish, which is a polynomial condition on the coefficients a_{ij}, b_{pq}. By equation (1.72), we see that this defines a polynomial condition in the affine coordinate system of $\varphi_{(1,\dots,d)} \times \varphi_{(1,\dots,d+1)}$, hence intersection $(*)$ is closed.

Now, for the second step, we use corollary (1.5.12) to conclude that

$$\begin{aligned} S_d = \mathrm{Gr}_1(V) \times \cdots \times \mathrm{Gr}_{d-1}(V) \times \{(F, F') \in \mathrm{Gr}_d(V) \times \mathrm{Gr}_{d+1}(V) : F \subset F'\} \times \cdots \\ \cdots \times \mathrm{Gr}_{d+2}(V) \times \cdots \times \mathrm{Gr}_n(V) \end{aligned} \tag{1.79}$$

is closed. Therefore

$$\mathcal{F}(V) = \bigcap_{d=1}^{n-1} S_d \tag{1.80}$$

is closed, so it's a projective variety.

Corollary 1.5.14. Since $\mathrm{Gr}_d(V)$ is covered by copies of intersecting affine planes, it's a irreducible variety.

Proposition 1.5.15. We have $\dim(\mathrm{Gr}_d(V)) = d(n-d),\ n = \dim(V)$.

Proof: From equation (1.73), we see that at $U_{(1,\dots,d)} \subset \mathbb{P}(\bigwedge^d V)$, the Grassmanian $\mathrm{Gr}_d(V)$ is defined by $V(I)$, where

$$I = (X_J - f_J(X) : \forall J = (j_1, \dots, j_d) \in \mathcal{J} \setminus \{(1, \dots, d), J_{pq} : p, q\}) \tag{1.81}$$

Put any LEX order on $k[X_J : J \in \mathcal{J} \setminus \{(1, \dots, d)\}]$, with the property $X_J > X_{J_{pq}}$ whenever $J \notin \{J_{pq} : (p,q) \in \{1, \dots, d\} \times \{d+1, \dots, n\}\}$. Then, we see that $\sqrt{I} = I$ and

$$\frac{k[X_J : J \neq (1, \dots, d)]}{I} \cong k[X_{J_{pq}} : (p,q) \in \{1, \dots, d\} \times \{d+1, \dots, n\}] \tag{1.82}$$

which has transcendence degree $d(n-d)$. □

Proposition 1.5.16. We have $\dim \mathcal{F}(V) = \frac{1}{2}n(n-1)$.

Proof: There is a simple way to represent an element $F \in \mathcal{F}(V)$. Fix a basis $\{e_1, \dots, e_n\}$ of V and observe that we can find a basis of V such that $F_i = \langle v_1, \dots, v_i \rangle$. Now, assume that $F \in U = U_{(1)} \times \cdots \times U_{(1,\dots,d-1)} \times U_{(1,\dots,d)}$. Then if we write $v_i = \sum_j a_{ij} e_j$, we see that $A = (a_{ij})$ has nonsingular principal minors $M_{ii},\ i = 1 \dots, n$. We call A a representative matrix of F.

Since $a_{11} \neq 0$, we can assume $v_1 = e_1 + \sum_{j>1} a_{1j} e_j$. Now, observe that if we only do elementary operations of the form $R_j \to R_j + kR_i,\ (i < j)$ and $R_j \to kR_j,\ (k \neq 0)$ (where

$R_i = (a_{i1}, \ldots, a_{in})$ is the ith row of A) on A, we still have A as a representative of the flag F. Using the hypothesis that the principal minors are nonsingular, we see that we can assume $a_{ii} = 1, \forall i$, while $a_{ij} = 0,\ (i > j)$. That is $v_i = e_i + \sum_{j>i} a_{ij} e_j$.

The above discussion shows that $\mathcal{F}(V) \cap U$ is defined by the graph of a morphism $c : k^{\frac{1}{2}n(n-1)} \to k^N$ for some N, so it's locally isomorphic to a plane $k^{\frac{1}{2}n(n-1)}$, which is an irreducible variety. Moreover, these affine planes cover $\mathcal{F}(V)$. This proves that $\mathcal{F}(V)$ is irreducible and that the dimension is $\frac{1}{2}n(n-1)$. □

Example 1.5.17. Suppose that $F \in \mathcal{F}(k^3)$ has matrix representation

$$A = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}. \tag{1.83}$$

Then, on the chart $U = U_{(1)} \times U_{(1,2)} \times U_{(1,2,3)}$ we have coordinates:

$$(\{(2) \mapsto a, (3) \mapsto b\}, \{(1,3) \mapsto c, (2,3) \mapsto ac - b\}, \{*\}) \tag{1.84}$$

Then we see that $U \cap \mathcal{F}(k^3)$ is the graph of the morphism $k^3 \to k : (a, b, c) \mapsto ac - b$.

1.6 Lecture 9

1.6.1 Induced function spaces

1.6.2 General definition of product

1.7 Lecture 10

In this section we let X to be an abstract variety and $Y \subset X$ to be an irreducible subset (a point is a case of interest). Define

$$\mathcal{O}_{X,Y} = \varinjlim_{Y \cap U \neq \varnothing} \mathcal{O}_X(U) \tag{1.85}$$

where the directed system is defined by $(\{\mathcal{O}_X(U)\}_{U \in \mathcal{U}}, \{\rho_{UV}\}_{U,V \in \mathcal{U}})$ where $\mathcal{U}$ is some system of generators for the topology of X. To avoid set theoretical issues we can take $\mathcal{U}$ to be the countable set of basic open sets of X.

Notation 1.7.1. We will denote the elements of $\mathcal{O}_{X,Y}$ by $\langle U, f \rangle$. So $\langle U, f \rangle = \langle V, g \rangle$ if and only if there exists $W \subset U \cap V$, $W \cap Y \neq \varnothing$ such that $f|_W = g|_W$. In particular, since Y is taken to be irreducible, we have that $W \cap Y \subset U \cap V \cap Y$ is dense, so $f|_{U \cap V \cap Y} = g|_{U \cap V \cap Y}$ by continuity of f and g.

Remark 1.7.2. If X is irreducible, we have an injection $\mathcal{O}_X(X) \hookrightarrow \mathcal{O}_{X,Y} : f \mapsto \langle X, f \rangle$. In fact, if $\langle X, f \rangle = 0$, then there exists $U \subset X, U \cap Y \neq \varnothing$ such that $f|_U = 0$. But since X is irreducible, $U \subset X$ is dense and thus $f = 0$.

Proposition 1.7.3. The ring $\mathcal{O}_{X,Y}$ is a k-algebra and is a local ring.

Proof: The inclusion

$$k \hookrightarrow \mathcal{O}_{X,Y} : c \mapsto \langle X, c\rangle \tag{1.86}$$

shows that $\mathcal{O}_{X,Y}$ is a $k-$algebra. Now, define the set

$$\mathfrak{m}_{X,Y} = \{\langle U, f\rangle : f|_{U\cap Y} = 0\}. \tag{1.87}$$

The condition $f|_{U\cap Y} = 0$ is well defined on the class $\langle U, f\rangle$. In fact, if $V \subset U$ is open, then $f|_{V\cap Y} = 0 \Rightarrow f|_{U\cap Y} = 0$, because $V \cap Y \subset U \cap Y$ is dense. Furthermore $\mathcal{O}_{X,Y}$ is clearly seen to be an ideal. Finally, we observe that

$$\mathcal{O}_{X,Y} - \mathfrak{m}_{X,Y} \subset \mathcal{O}^*_{X,Y}. \tag{1.88}$$

In fact, if $\langle U, f\rangle$ is such that $f(U \cap Y) \neq \{0\}$, then the set $V = \{q \in U : f(q) \neq 0\}$ is a nonempty open set of U. Therefore

$$\langle U, f\rangle = \langle V, f|_V\rangle = \langle V, \frac{1}{f}\Big|_V\rangle^{-1} \in \mathcal{O}^*_{X,Y}. \tag{1.89}$$

□

Proposition 1.7.4. Let $X \subset k^m$ be an affine algebraic set and $Y \subset X$ be an irreducible subset. Denote by $P = I(Y)/I(X) \subset A = k[X]$, then P is a prime ideal and we have an isomorphism

$$\psi : A_P \to \mathcal{O}_{X,Y} : \overline{F}/\overline{G} \mapsto \langle D_X(G), F/G\rangle. \tag{1.90}$$

Proof: First observe that $G \in I(Y) \iff G|_Y = 0 \iff D_Y(G) = \varnothing$. Therefore $\overline{G} \notin P \iff Y \cap D(G) \neq \varnothing$. Furthermore, we have $\overline{F}_1/\overline{G}_1 = \overline{F}_2/\overline{G}_2$ in A_P if and only if there exists $G \notin P$ such that

$$\begin{aligned}
&\overline{G}(\overline{F}_1\overline{G}_2 - \overline{F}_2\overline{G}_1) = 0 \in k[X] \iff \\
&G(F_1G_2 - F_2G_1)|_X = 0 \in \mathcal{O}_X(X) \iff \\
&(F_1G_2 - F_2G_1)|_{D_X(G)} = 0 \in \mathcal{O}_X(D_X(G)) \Rightarrow \\
&\langle D_X(G), F_1/G_1\rangle = \langle D_X(G), F_2/G_2\rangle.
\end{aligned} \tag{1.91}$$

Then we see that ψ is well defined. To see injectivity, if $\langle D_X(G), F/G\rangle = 0$, then $F|_U = 0$ for some open set U such that $U \cap Y \neq \varnothing$. But since $U \subset X$ is open dense, we have $F|_X = 0 \iff \overline{F} = 0$.

It remains to prove that it's surjective. To see this, let $\langle U, f\rangle \in \mathcal{O}_{X,Y}$, and let $q \in U \cap Y$. Since $f : U \to k$ is regular, there exists $U' \ni q, U' \subset U$ and polynomials $F, G \in k[\underline{X}], G(q') \neq 0\ \forall q' \in U'$ such that $f|_{U'} = F/G|_{U'}$. In particular, $q \in D_X(G) \cap Y$ and $\langle U, f\rangle = \langle D_X(G), F/G\rangle$. To see that P is prime, we observe that it's the image of the prime ideal $I(Y)$ under the surjective morphism $f : k[X_1, \ldots, X_m] \to k[X]$ and $P \subset \ker(f)$. □

Example 1.7.5. Let $X = k^1$ and $Y = \{0\}$, we have $A = k[t]$ and $P = I(Y) = (t)$. So we see that $A_P = k[t]_P = \{F/G : G \notin (t)\}$. For the other side, we have

$$\mathcal{O}_{X,Y} = \{\langle D(G), F/G\rangle : G(0) \neq 0\} \tag{1.92}$$

because $D(G) \cap Y \neq \varnothing \iff G(0) \neq 0$.

1.8 Lecture 11

1.8.1 Birational geometry

1.9 Lecture 12

TODO: Describe the variety structure of $k^n \times \mathbb{P}^m$ explictly.

1.9.1 Blowup of affine varieties

Consider the following data.

- $X \subset k^n$ an affine variety, $A = \mathcal{O}_X(X)$ its coordinate ring;
- $I \subset A$ an ideal such that if $Y = V(I)$, then $U = X - Y \subset X$ is dense.

Denote $I = (f_1, \ldots, f_m)$, $f_i = F_i|_X$ and

$$h : k[X_1, \ldots, X_n, Y_1, \ldots, Y_m] \to A[T] : X_i \mapsto x_i = X_i|_X, Y_i \mapsto f_i T. \tag{1.93}$$

Let $J = \mathrm{Ker}(h)$, then h is homogeneous in $\underline{Y}$. In fact, suppose that $F = F_0 + F_2 + \cdots \in J$, where F_i the ith degree homogeneous component of F viewed as a polynomial in $k[\underline{X}][\underline{Y}]$. Then

$$h(F_d) = F_d(x_1, \ldots, x_n, f_1 T, \ldots, f_m T) = F_d(x_1, \ldots, x_n, f_1, \ldots, f_m) T^d \in A[T]. \tag{1.94}$$

So $h(F) = 0 \Rightarrow h(F_d) = 0$ and $F_d \in J$, $\forall d \geq 0$.

Remark 1.9.1. An important observation is that $F_i Y_j - F_j Y_i \in J, \forall i, j$ by construction.

Now we can define

$$\widetilde{X} = V_{a|p}(J) \subset X \times \mathbb{P}^{m-1} \subset k^n \times \mathbb{P}^{m-1}. \tag{1.95}$$

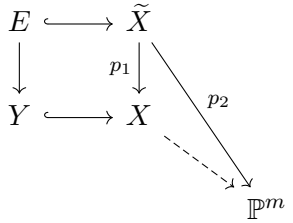
Observe that the first insclusion is a consequence of $p_1(\widetilde{X}) \subset X$, where p_1 is the projection in $\underline{X}$. In fact, we have a injective map $p_1^{\#} : k[\underline{X}] \hookrightarrow k[\underline{X}, \underline{Y}] : g \mapsto g \circ p_1$. Therefore $g \circ p_1 \in J \iff g(x_1, \ldots, x_n) = 0 \iff g|_X = 0$, so $p_1^{\#}(I(X)) = J$ and $(a, b) \in \widetilde{X}$ implies that $g(a) = 0$, $\forall g \in I(X)$, i.e., $a \in VI(X) = X$.

Remark 1.9.2. By the discussion above, we see that $p_1^{\#}$ descends to a map $k[X] \to k[\underline{X}, \underline{Y}]/J$.

Remark 1.9.3. Denote by $U_i = \widetilde{X} \cap D(Y_i)$. Observe that $D(Y_i) = k^n \times D_+(Y_i) \subset k^n \times \mathbb{P}^m$ it's a open affine subset. So $U_i \subset D(Y_i)$ is closed, because it's the intersection of $D(Y_i)$ with a closed subset $\widetilde{X} \subset k^n \times \mathbb{P}^m$. As a closed subset of a affine subset, U_i is an affine subset. In U_i we have the relation $F_j = (\frac{Y_j}{Y_i})F_i$. Therefore

$$p_1^{\#}(I)\mathcal{O}_{\widetilde{X}}(U_i) = (f_i \circ p_1|_{U_i}). \tag{1.96}$$

Now we define $E = p_1^{-1}(Y) = V_{a|p}(f_1 \circ p_1, \ldots, f_m \circ p_1)$ so we have the diagram.



The dashed arrow is the map $p \mapsto (f_1(p) : \cdots : f_m(p))$, it's not defined only on Y and since $X - Y$ is dense in X by hypothesis, it's a rational map.

Definition 1.9.4. The Rees algebra of $I \subset A$ is defined as the image the map h defined below. That is, we let

$$\mathrm{Rees}_I(A) = A \oplus IT \oplus I^2T^2 \oplus \cdots \oplus I^dT^d \oplus \cdots \subset A[T]. \tag{1.97}$$

Remark 1.9.5. Remember that $I = \bigcap_{P \supset I,\ \text{prime}} P$. We say that A is reduced if $\sqrt{(0)} = (0)$, i.e, $x^n = 0 \Rightarrow x = 0$. In particular, this means that $(0) = \bigcap_{P\ \text{prime}} P$. Now, assume that A is a reduced ring (it's the case of $A = \mathcal{O}_X(X)$), we claim that $A[T]$ is reduced. In fact, $f^n = 0 \in A[T]$ implies that $f^n = 0 \in (\frac{A}{P})[T]$ for all P prime. Since A/P is a domain, we see that $f = 0 \in (\frac{A}{P})[T]$ for any prime P. So if $f = a_0 + a_1T + \cdots$, we get that $a_i \in P$ for all prime P for all $i \geq 0$. Therefore $a_i = 0, ; \forall i$ because A is reduced.

By the previous remark, we see that $\mathrm{Rees}_I(A)$ is reduced and since $k[\underline{X}, \underline{Y}]/J \cong \mathrm{Rees}_I(A)$, we get that J is radical ($J = \sqrt{J}$).

Lemma 1.9.6. If $Z \subset k^m$ is affine and $f \in \mathcal{O}_Z(Z)$ is such that f is not a divisor of zero in $\mathcal{O}_Z(Z)$ if and only if $Z - V(f) \subset Z$ is dense.

Proof: It's not dense if and only if $D_Z(g) \subset V(f)$ for some g, if and only if $fg|_Z = 0 \Rightarrow f|_Z$ is a divisor of zero. □

Furthermore, we have that $X - Y = X - V(f_1, \ldots, f_m) \subset X$ is dense. So, if $gf_i|_X = 0,\ \forall i$, then $D_X(g) \subset V(f_1, \ldots, f_m)$, so $X - Y$ is not dense, which is a contradiciton. Then we see that there exists $i,\ f_i \nmid 0$.

Proposition 1.9.7. $\forall i,\ f_i \circ p_1|_{U_i}$ is not a divisor of zero.

Proof: Let $b \in \mathcal{O}_{\widetilde{X}}(U_i) \setminus \{0\}$ such that $b \cdot f_i \circ p_1|_{U_i} = 0$ (by contradiction). Since U_i is affine (see remark (1.9.3)), we have $b = B/Y_i^d|_{U_i}$ (justify this !) with $B \in k[\underline{X}, \underline{Y}]$ homogeneous of degree d in Y. Since $b \neq 0$, we have $B \notin J$. Now

$$\frac{B}{Y_i^d} \cdot F_i\Big|_{\widetilde{X} \cap D(Y_i)} = 0 \Rightarrow BF_iY_i|_{\widetilde{X}} = 0 \tag{1.98}$$

which implies that $BF_iY_i \in \sqrt{J} = J$. Therefore

$$\begin{aligned} & B(x_1, \ldots, x_n, f_1T, \ldots, f_mT)f_i^2T = 0 \\ & \Rightarrow B(x_1, \ldots, x_n, f_1, \ldots, f_m)f_i^2T = 0 \\ & \Rightarrow B(x_1, \ldots, x_n, f_1, \ldots, f_m)|_{X-Y} = 0 \quad (\textit{why?}) \\ & \Rightarrow B(x_1, \ldots, x_n, f_1, \ldots, f_m) = 0 \\ & \Rightarrow B(X_1, \ldots, X_n, Y_1, \ldots, Y_m) \in J \end{aligned} \tag{1.99}$$

which is a contradiction. □

Proposition 1.9.8. $\widetilde{X} - E \subset \widetilde{X}$ is dense.

Proof: By the last proposition, we see that $U_i - E \cap U_i = U_i - V(f_i \circ p_1)$ is dense in U_i and since $E \subset \widetilde{X} = \bigcup_i U_i$, we get that $\widetilde{X} - E$ is dense in $\widetilde{X}$. □

Summary: We constructed the following data.

$$\begin{array}{ccccc} E = b^{-1}(Y) & \hookrightarrow & \widetilde{X} & \xleftarrow{\text{dense}} & \widetilde{X} - E \\ \downarrow & & \downarrow{\scriptstyle b} & & \downarrow{\scriptstyle s} \\ Y & \hookrightarrow & X & \xleftarrow[\text{dense}]{} & X - Y \end{array}$$

Furthermore, there exists a covering $\widetilde{X} = \bigcup_i U_i$ of open affines, sections $b_i \in \mathcal{O}_X(U_i),\ b_i \nmid 0$ such that $I\mathcal{O}_X(U_i) = (b_i)$.

1.9.2 universal property

1.9.3 Examples

1.9.4 Generalization

1.10 Lecture 13

1.10.1 Dimension

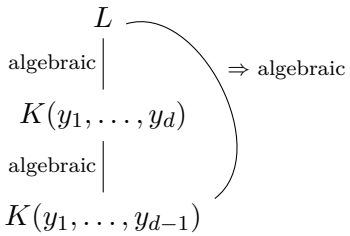
Definition 1.10.1. (Transcendence degree). Let $L|K$ be a field extension, we define its transcendence degree by

$$\begin{aligned}\text{Tr.deg}(L|K) &= \min\{d : \exists\, y_1, \dots, y_d \in L \text{ such that } L|K(y_1, \dots, y_d) \text{ is algebraic}\} \\ &= \max\{d : \exists\, y_1, \dots, y_d \in L \text{ such that } y_1, \dots, y_d \text{ are A.I over } K\}\end{aligned} \tag{1.100}$$

To prove the equality in the above definition, let d be the maximum positive integer such that $y_1, \dots, y_d$ is algebraically independent (A.I) over K. Then, for any $z \in L$, the set $\{y_1, \dots, y_d, z\}$ is algebraically dependent over K, so there exists $G \in K[T_1, \dots, T_{d+1}] \backslash 0$ such that $G(y_1, \dots, y_d, z) = 0$. This is equivalent to say that z is algebraic over $K(y_1, \dots, y_d)$, so $L|K(y_1, \dots, y_d)$ is algebraic. Then we get

$$d_{\min} \leq d_{\max}. \tag{1.101}$$

Another way to see this inequality is as follows. Let d be the minimum positive integer such that $L|K(y_1, \dots, y_d)$ is algebraic and suppose that there exists $G \in K[T_1, \dots, T_d] \setminus 0$ such that $G(y_1, \dots, y_d) = 0$. Without lost of generality, we assume that $\deg_{T_d}(G) > 0$. But this means that



which is a contradiction, by minimality of d.TODO: prove the other inequality

Remark 1.10.2. If R is a k-algebra and a integral domain. We define its transcendence degree over k by

$$\text{Tr.deg}(R) = \text{Tr.deg}(\text{Frac}(R)|k). \tag{1.102}$$

Definition 1.10.3. Let X be an irreducible affine variety. We define its dimension by

$$\dim(X) = \text{Tr.deg}(K(X)|k) \tag{1.103}$$

Remark 1.10.4. We see that open affine subsets behave as we expect:

$$\dim(U) = \text{Tr.deg}(K(U)) = \text{Tr.deg}(X) = \dim(X) \tag{1.104}$$

because $K(X) \cong K(U)$, whenever X is irreducible.

Remark 1.10.5. Let R be a k-algebra and a integral domain. By the Noether normalization lemma, there exists $x_1, \ldots, x_d$ algebraically independent (A.I) such that R is integral over $k[x_1, \ldots, x_d]$, which implies that $\text{Frac}(R)|k(x_1, \ldots, x_d)$ is algebraic. Then we see that

$$\text{Tr.deg}(R|k) \leq d \leq \text{Tr.deg}(R|k) \quad \text{(see definition (1.10.1))} \tag{1.105}$$

so $\text{Tr.deg}(\text{Frac}(R)|k) = d$.

Proposition 1.10.6. Let $X = V(F) \subset k^n$ be an irreducible affine variety, then $\dim X = n - 1$. Algebraically this means that

$$\text{Tr.deg}(\frac{k[X_1, \ldots, X_n]}{(F)}) = n - 1. \tag{1.106}$$

Proof: Since X is irreducible, then F is irreducible. Otherwise $F = GH \Rightarrow V(F) = V(G) \cup V(H)$, which is not irreducible. Without lost of generality, suppose that

$$\deg_{X_n} F(X_1, \ldots, X_n) > 0. \tag{1.107}$$

Write $x_i = X_i + (F) \in \mathcal{O}_X(X) = k[\underline{X}]/(F)$, x_n is algebraic over $k[x_1, \ldots, x_{n-1}]$. Now, suppose that $G \in k[T_1, \ldots, T_{n-1}]$ is such that $G(x_1, \ldots, x_{n-1}) = 0$, then

$$G(X_1, \ldots, X_{n-1})|_X = 0 \iff G(X_1, \ldots, X_{n-1}) \in (F) \tag{1.108}$$

and $G = 0$. So $\{x_1, \ldots, x_{n-1}\}$ are algebraically independent and $\dim(X) \geq n-1$. However, the extension $K(X)|k(x_1, \ldots, x_{n-1})$ is algebraic because

$$K(X) = k(x_1, \ldots, x_{n-1})(x_n). \tag{1.109}$$

So $\dim(X) \leq n - 1$ and we see that we have the equality. □

Proposition 1.10.7. Let X be an irreducible variety and $Y \subset X$ a proper closed irreducible subset. Then $\dim(Y) < \dim(X)$.

Proof: Let $U \subset X$ be an open affine such that $U \cap Y \neq \varnothing$. Then

$$\dim U = \dim X,\ \dim U \cap Y = \dim Y. \tag{1.110}$$

So we can assume X affine irreducible. Write $R = \mathcal{O}_X(X)$ and let $I = I(Y) \subset R$. If $I = (0)$, then $Y = VI = V(0) = X$ which is a contradiction. So, there exists $f \in I \setminus 0$. By Noether normalization lemma, there exists $x_1, \ldots, x_d \in R$ (A.I) such that $R \supset k[x_1, \ldots, x_d]$ is integral. Now $x_1, \ldots, x_d, f$ must be algebraically dependent, so $G(x_1, \ldots, x_d, f) = 0$ for some $G \in k[T_1, \ldots, T_{d+1}] \setminus 0,\ T_{d+1} \nmid G$. Now, on $\mathcal{O}_Y(Y) = R/I$, we have

$$\begin{aligned} G(x_1, \ldots, x_d, 0) + I &= G(x_1 + I, \ldots, x_d + I, I) \\ &= G(x_1 + I, \ldots, x_d + I, f + I) \\ &= G(x_1, \ldots, x_d, f) + I \\ &= I. \end{aligned} \tag{1.111}$$

Therefore $x_1 + I, \ldots, x_d + I$ are algebraically dependent over k. However $\mathcal{O}_Y(Y)$ is integral over $k[x_1 + I, \ldots, x_d + I]$, so $K(Y)|k(x_1 + I, \ldots, x_d + I)$ is algebraic and $\dim(Y) \leq d$. The equality can not happen, because $\{x_i + I\}$ is A.D, so $\dim(Y) < d$. □

Theorem 1.10.8. Let X be an irreducible affine variety, $f : X \to k$ regular ($f \neq 0$). Let $Z \subset V(f)$ be a irreducible component. Then

$$\dim Z = \dim X - 1. \tag{1.112}$$

Proof: Write $R = \mathcal{O}_X(X)$. Let

$$V(f) = Z_1 \cup \cdots \cup Z_m,\ Z = Z_1. \tag{1.113}$$

Take $x \in Z \setminus \bigcup_{j>1} Z_j$, then there exists an open affine $U \subset X - Z_2 \cup \cdots \cup Z_m, x \in U$, i.e., there exists an open affine U such that $U \cap Z \neq \varnothing$ and $U \cap (Z_2 \cup \cdots \cup Z_m) = \varnothing$. In this condition

$$V(f_U) = V(f) \cap U = Z \cap U \text{ is irreducible.} \tag{1.114}$$

Therefore

$$\dim X = \dim U,\ \dim V(f) \cap U = \dim Z \cap U \overset{Z \text{ irred.}}{=} \dim Z. \tag{1.115}$$

Therefore, we want to prove that

$$\dim U = \dim V(f) \cap U + 1 = \dim V(f|_U) + 1. \tag{1.116}$$

Then we assume $Z = V(f)$ to be an irreducible affine variety, so $I(Z) = \sqrt{(f)}$ is prime.

By Noether normalization lemma, there exists $x_1, \ldots, x_d \in R$ such that R is integral over $S = k[x_1, \ldots, x_d]$. Therefore we have the diagram

$$\begin{array}{ccc} R & \hookrightarrow & K = K(X) \\ \uparrow & & \uparrow \text{finite} \\ S & \hookrightarrow & L \end{array}$$

where $L = \mathrm{Frac}(S)$. We note that $K|L$ is algebraic and finitely generated. In fact, if $X \subset k^n$, then $K(X) = k(\overline{x}_1, \ldots, \overline{x}_n) = L(\overline{x}_1, \ldots, \overline{x}_n)$. Therefore $K|L$ is a finite extension. Now, let $P(T) \in L(T)$ be the minimum polynomial of f over L and write

$$P(T) = T^m + a_1 T^{m-1} + \cdots + a_m. \tag{1.117}$$

Since f is integral over S (which is a UFD), its minimum polynomial has coefficients in S, i.e., $a_i \in S$, $\forall i$. Now define $f_0 = N_{K|L}(f)$. We have

$$\begin{aligned} f_0 &= N_{K|L}(f) \\ &= N_{L(f)|L} N_{K|L(f)}(f) \\ &= N_{L(f)|L}\Big(\prod_{\sigma \in \mathrm{Hom}_{L(f)}(K, \overline{k})} \sigma(f) \Big) \\ &= N_{L(f)|L}(f^{\#\mathrm{Hom}_{L(f)}(K,\overline{k})}) \\ &= N_{L(f)|L}(f^{\#[K:L(f)]}) \\ &= N_{L(f)|L}(f)^{\#[K:L(f)]}. \end{aligned} \tag{1.118}$$

Write $p = \#[K : L(f)]$, so $f_0 = \pm a_m^p \in S$.

Write $P = \sqrt{(f)} \subset R$. We draw the diagram

$$\begin{array}{ccccc} P & \hookrightarrow & R & \hookrightarrow & K \\ \uparrow & & \uparrow & & \uparrow \text{finite} \\ P \cap S & \hookrightarrow & S & \hookrightarrow & L \end{array}$$

We claim that $P \cap S = (f_0)$. In fact, we have

$$(f^m + a_1 f^{m-1} + \cdots + a_{m-1} f)^p = \pm a_m^p = \pm f_0. \tag{1.119}$$

Therefore $f_0 \in fR \subset P \cap S$. Now, if $g \in P \cap S$, then $g^n \in (f)$ and $g^n = fh$ for some $h \in R$. Taking the norm we get

$$g^{n[K:L]} = N_{K|L}(g^n) = N_{K|L}(f) N_{K|L}(h) = f_0 N_{K|L}(h) \in (f_0). \tag{1.120}$$

Therefore $g \in \sqrt{f_0}$ and $P \cap S \subset (f_0)$. Now we obtain

$$\begin{array}{c} R/P = \mathcal{O}_Z(Z) \\ \Big| \text{integral} \\ S/(P \cap S) = S/\sqrt{(f_0)} \end{array}$$

Finally $\text{Tr.deg}(S/\sqrt{(f_0)}) = d-1$ (see remark below), so there exists $y_1, \dots, y_{d-1}$ (A.I) such that

$$\begin{array}{c} R/P = \mathcal{O}_Z(Z) \\ \Big| \text{integral} \\ S/(P \cap S) \\ \Big| \text{integral} \\ k[y_1, \dots, y_{d-1}] \end{array}$$

This means that $K(Z)|k(y_1, \dots, y_{d-1})$ is algebraic with $\{y_i\}_{i=1}^{d-1}$ A.I. Therefore $\dim Z = d-1$. □

Remark 1.10.9. In the previous proof we had the equality $\text{Tr.deg}(S/\sqrt{(f_0)}) = d-1$. First, remember that $S = k[x_1, \dots, x_d]$, where $\{x_i\}_{i=1}^d$ are algebraically independent, so it's a polynomial ring. Now, let $V = V(f_0)$, then $I(V) = \sqrt{(f_0)}$ by Hilbert Nullstelensatz. Therefore $S/\sqrt{(f_0)} = \mathcal{O}_V(V)$ is the coordinate ring of a hypersurface of k^d and it has transcendence degree $d-1$ (see proposition (1.10.6)).

1.10.2 Topological interpretation of dimension

Lemma 1.10.10. Let $Y \subset X$ be a maximal irreducible proper closed subset, where X is a irreducible variety. Then $\dim(X) = \dim(Y) + 1$.

Proof: Let $U \subset X$ be an open affine subset such that $U \cap Y \neq \varnothing$. Then

$$\dim X = \dim U, \ \dim Y = \dim Y \cap U. \tag{1.121}$$

Furthermore, $Y \cap U \neq U$, otherwise we would have

$$Y = Y \cap \overline{U} \supset \overline{Y \cap U} = \overline{U} = X \tag{1.122}$$

and $X = Y$. Also if $Y \cap U \subset Z$, where $Z \subset U$ is a closed irreducible subset, then we have

$$Y \cap \overline{U} \subset Z \Rightarrow Y \subset Z \tag{1.123}$$

and since Y is maximal, we have $Z = Y$ or $Z = X \cap U = U$. Then we see that $Y \cap U$ is a maximal proper irreducible subset of U. Then we can assume X affine and Y an irreducible component of X. In particular if $f \in I(Y) \setminus 0$, then $Y \subset V(f)$ is maximal irreducible, so it's an irreducible component. Then we apply theorem (1.10.8). □

Proposition 1.10.11. Let X be an irreducible variety. Then for any chain of irreducible closed subsets

$$\varnothing \neq Z_0 \subset Z_1 \subset \cdots \subset Z_r \subset X,\ Z_{i-1} \neq Z_i\ (i = 1, \ldots, r) \tag{1.124}$$

we have $r \leq \dim X$ with equality if and only if the chain is maximal.

Proof: By proposition (1.10.7), we have $\dim(Z_{i-1}) < \dim(Z_i),\ i = 1, \ldots, r$. This implies that

$$r \leq \sum_{i=1}^{r} (\dim(Z_i) - \dim(Z_{i-1})) = \dim(Z_r) - \dim(Z_0) \leq \dim(X). \tag{1.125}$$

Now, if the chain is maximal then for each $i,\ i = 1, \ldots r$ we have $Z_{i-1} \subset Z_i$ a maximal proper irreducible subset. In this case we must have $\dim(Z_{i-1}) = \dim(Z_i) - 1$, because of the previous lemma. Furthermore, in this case $Z_r = X$ and Z_0 is a point. Therefore

$$r = \sum_{i=1}^{r} (\dim(Z_i) - \dim(Z_{i-1})) = \dim(X). \tag{1.126}$$

□

These results gives motivatio to the following definition.

Definition 1.10.12. Let X be a topological space. Then we define

$$\dim X = \sup\{r : \exists\ Y_0 \subset \cdots \subset Y_r \subset X,\ Y_{i-1} \neq Y_i \text{ irreducible}\} \tag{1.127}$$

Proposition 1.10.13. Let X be a variety, we have $\dim X = \max_i \dim X_i$ where X_i are the irreducible components of X.

1.11 Lecture 14 and 15

1.11.1 Fibers of a morphism

Definition 1.11.1. A morphism $f : X \to Y$ is said to be dominant if $\overline{f(X)} = Y$.

Proposition 1.11.2. Let $f : X \to Y$ be a morphism of algebraic varieties and $Z = \overline{f(X)} \subset Y$. If X is irreducible, then Z is irreducible and $f^{\#}$ induces a injective morphism $K(Z) \hookrightarrow K(X)$, so $\dim(Z) \leq \dim(X)$.

Proof: Suppose that X is irreducible and let $Z = Z_1 \cup Z_2$ with $Z_i,\ i = 1, 2$ closed. Then $X = f^{-1}(Z_1) \cup f^{-1}(Z_2)$, so $X = f^{-1}(Z_1)$ say. So we get $f(X) \subset Z_1$ and $Z = \overline{f(X)} \subset Z_1$ and $Z = Z_1$. This proves that Z is irreducible. Now we have an induced morphism $f : X \to Z$ which is dominant. Let $s \in \mathcal{O}_Z(Z)$, then $f^{\#}(s) = 0 \Rightarrow s = 0$ on a dense subset of Z, so it must be zero. This proves that $f^{\#}$ induces an embedding $K(Z) \hookrightarrow K(X)$. □

Remark 1.11.3. It's useful to remember that whenever we have a chain of proper closed subsets $X \subset Y \subset Z$, then $I(Y) \subset I(X)$ is a proper subset. If not, we have $X = VI(X) = VI(Y) = Y$. Here we write $I(X) = \{f \in \mathcal{O}(Z) : f|_X = 0\}$.

Lemma 1.11.4. Let Y be an irreducible affine algebraic variety and $W \subset Y$ a proper, closed and irreducible subset. Let $s = \dim(Y) - \dim(W)$, then there exists $f_1, \ldots, f_s \in \mathcal{O}(Y)$ such that W is an irreducible component of $V(f_1, \ldots, f_s)$ and every irreducible component of $V(f_1, \ldots, f_s)$ has codimension equal to s.

Proof: First we claim that there exists a chain if closed irreducible subsets

$$W = Z_s \subset Z_{s-1} \subset \cdots \subset Z_1 \subset Z_0 = Y \tag{1.128}$$

such that $\operatorname{codim}(Z_i, Y) = i$. To see this, let $f \in I(W) \setminus 0$, then $W \subset V(f) \subset Y$. Since W is irreducible, it's contained in some irreducible component Z_1 of $V(f)$, which has codimension 1 in Y. Now, repeat this step with $Y = Z_1$. It remains to prove that in the sth step, we have $W = Z_s$. If this doesnt happen, then $\dim W < \dim Z_s$ and $\operatorname{codim}(W, Y) > s$, which is a contradiction.

Now, we claim that there exists a collection of regular functions $f_i \in \mathcal{O}(Z),\ i = 1, \ldots, s$ such that for every $i = 1, \ldots, s$, the set Z_i is an irreducible component of $V(f_1, \ldots, f_i)$ and every irreducible component of $V(f_1, \ldots, f_i)$ has codimenion i in Z. For $i = 1$, it's sufficient to take $f_1 \in I(Z_1) \setminus 0$, so $Z_1 \subset V(f_1)$ is an irreducible component, because it has codimenion one in Y, and every other irreducible component has codimenion one because of theorem (1.10.8).

Now, we want to find f_2 such that $Z_2 \subset V(f_1, f_2)$ is an irreducible component of codimension 2, while every other irreducible component has codimension 2 in Y. To do this, let $Z_1 = X_1, \ldots, X_m$ be the irreducible components of $V(f_1)$. Then $Z_2 \subset Z_1$ and $I(Z_2) \neq I(X_j)$ for every j, because $\operatorname{codim}(Z_2, Y) > \operatorname{codim}(X_j, Y)$. Therefore, by prime avoidance theorem, we know that $I(Z_2) \neq \bigcup_j I(X_j)$ and there exists

$$f_2 \in I(Z_2) - \bigcup_j I(X_j) \tag{1.129}$$

not zero. This is our candidate. In fact, we have $Z_2 \in V(f_1, f_2)$, which we show to be an irreducible component. We have $V(f_1, f_2) = \bigcup_j X_j \cap V(f_2) = \bigcup_j V(f_2|_{X_2})$ and since Z_2 is irreducible, it must be contained in some irreducible component of $V(f_2|_{X_j})$. However, any irreducible component of $V(f_2|_{X_j}) \subset X_j$ has irreducible component of codimension 1 in X_j

(see (1.10.8)), thus it has codimension 2 in Y. Therefore Z_2 is an irreducible component of $V(f_2|_{X_j})$ by dimensional reasons. Since the collection of irreducible components of $V(f_1, f_2)$ is the union of the irreducible components of $V(f_2|_{X_j})$ for each j, we conclude that each irreducible component of $V(f_1, f_2)$ has codimension 2 in Y. The induction step follows this pattern. □

Theorem 1.11.5. Let $f : X \to Y$ be a dominant morphism of irreducible affine varieties. Let $W \subset Y$ be an irreducible closed subset and $Z \subset f^{-1}(W)$ be an irreducible component such that $\overline{f(Z)} = W$ (this is an additional hypothesis). Then

$$\operatorname{codim}(Z, X) \leq \operatorname{codim}(W, Y). \tag{1.130}$$

A picture to remember:

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ \uparrow & & \uparrow \\ Z \subset f^{-1}(W) & \xrightarrow[f]{} & W \end{array}$$

Proof: Let $U \subset Y$ be an open affine such that $U \cap W \neq \varnothing$. Then $f(Z) \cap U \neq \varnothing$. Then we get the diagram

$$\begin{array}{ccc} Z \cap f^{-1}(W \cap U) & \overset{\text{irr. comp.}}{\hookrightarrow} & f^{-1}(W \cap U) \\ \downarrow & & \downarrow \\ Z & \xhookrightarrow[\text{irr. comp.}]{} & f^{-1}(W) \end{array}$$

Furthermore, we have

$$\dim U = \dim Y, \dim W \cap U = \dim W, \dim(Z \cap f^{-1}(W \cap U)) = \dim Z \tag{1.131}$$

and $\dim f^{-1}(U) = \dim X$. Then it's sucfficient to prove the result for $f|_{f^{-1}(U)} : f^{-1}(U) \to U$ and $W \cap U \subset U$ irreducible closed. So we assume that Y is an affine irreducible algebraic variety. Let $s = \dim Y - \dim W$, then by the previous lemma, there exists $f_1, \ldots, f_s \in \mathcal{O}(Y)$ such that W is irreducible component of $V(f_1, \ldots, f_s)$. Let $g_i = f_i \circ f \in \mathcal{O}(X)$. Then $Z \subset f^{-1}(W) = V(g_1, \ldots, g_s)$ is an irreducible component. In fact, if $Z \subset Z'$, where $Z' \subset V(g_1, \ldots, g_s)$ is an irreducible component, then $W = \overline{f(Z)} \subset \overline{f(Z')} \subset V(f_1, \ldots, f_s)$ and $W = \overline{f(Z')}$, because $\overline{f(Z')}$ is irreducible. This implies that $f(Z') \subset W \Rightarrow Z' \subset f^{-1}(W)$ and since Z is a irreducible component, we have $Z = Z'$. Finally, since $Z \subset V(g_1, \ldots, g_s)$ is an irreducible component, we must have $\operatorname{codim}(Z, X) \leq s$. □

Remark 1.11.6. Let X be an irreducible affine algebraic variety and $f_1, \ldots, f_s \in \mathcal{O}(X)$. Then if $Z \subset V(f_1, \ldots, f_s)$ is an irreducible component, we claim that $\operatorname{codim}(Z, X) \leq s$. We

apply induction on s, the base case being just theorem (1.10.8). Let $Z' \subset V(f_1, \ldots, f_{s-1})$ be an irreducible component containing Z, then by induction, we have $\operatorname{codim}(Z', X) \leq s-1$. Now we have

$$Z \subset V(f_s|_{Z'}) \subset V(f_1, \ldots, f_s). \tag{1.132}$$

It's important to note that if $Z'' \subset V(f_s|_{Z'})$ is an irreducible component containing Z, then $Z \subset Z'' \subset V(f_1, \ldots, f_s)$ and $Z = Z''$ because Z is an irreducible component of $V(f_1, \ldots, f_s)$, so it's maximal irreducible. Then Z is an irreducible component of $V(f_s|_{Z'})$. If $f_s|_{Z'} = 0$, then $Z \subset Z'$ is an irreducible component and $Z = Z'$. So $\operatorname{codim}(Z, X) = \operatorname{codim}(Z', X) \leq s-1 \leq s$. If $f_s|_{Z'} \neq 0$, then $\dim Z = \dim Z' - 1$ by theorem (1.10.8). Therefore $\operatorname{codim}(Z, X) = \operatorname{codim}(Z', X) + 1 \leq s$.

Corollary 1.11.7. If $W = \{y\}$ and $Z \subset f^{-1}(W)$ is an irreducible component, then $\dim Z \geq \dim X - \dim Y$.

Lemma 1.11.8. Let Z be a topological space such that $Z = \bigcup_i U_i$ is a covering of open irreducible subsets. If $U_i \cap U_j \neq \varnothing$, $\forall i, j$, then Z is irreducible.

Proof: Let U, V be two nonempty open subsets. Then $U \cap U_i \neq \varnothing, V \cap U_j \neq \varnothing$ for some i, j. But since $U \cap U_i, U_i \cap U_j \subset U_i$ are two open subsets, we see that $U \cap U_i \cap U_j \neq \varnothing$ because U_i is irreducible. In a similar way, we have $V \cap U_i \cap U_j \neq \varnothing$. Furthermore $U_i \cap U_j \subset U_i$ is an open subset of an irreducible one, thus it is irreducible. Therefore $U \cap V \cap U_i \cap U_j = (U \cap U_i \cap U_j) \cap (V \cap U_i \cap U_j) \neq \varnothing$ and $U \cap V \neq \varnothing$. □

Proposition 1.11.9. Let X, Y be two irreducible algebraic varieties, then $X \times Y$ is irreducible.

Proof: Let $U \subset X, V \subset Y$ be two open affine subsets. Then if we show that $U \times V$ is irreducible, we conclude that $X \times Y$ is irreducible by the lemma above. So we assume X, Y affine irreducible varieties. Suppose that $X \times Y = Z_1 \cup Z_2$, then for each $y \in Y$ we have

$$X \times \{y\} \subset Z_1 \text{ or } X \times \{y\} \subset Z_2 \tag{1.133}$$

because $X \times \{y\}$ is irreducible. Therefore $Y = Y_1 \cup Y_2$, where

$$Y_i = \{y \in Y : X \times \{y\} \subset Z_i\}. \tag{1.134}$$

However, if $y \in Y - Y_1$, then there exists $x \in X$ such that $(x, y) \notin Z_1$. However there exists $U \cap Y$ such that $\{x\} \times U = \{x\} \times Y - (\{x\} \times Y) \cap Z_1 \neq \varnothing$, so $y \in U \subset Y - Y_1$. This proves that Y_1, Y_2 are closed, so $Y = Y_1$ say. Then $X \times Y \subset Z_1 \Rightarrow X \times Y = Z_1$. □

Remark 1.11.10. The projection $\pi|_{\{x\}\times Y} : \{x\} \times Y \xrightarrow{\cong} Y$ is a continuous bijection. But we have more, it's a open map, thus a homeomorphism. In fact, we have

$$\{x\} \times D_Y(g) = (\{x\} \times Y) \cap D_{X\times Y}(gp_2), g \in \mathcal{O}(Y). \tag{1.135}$$

Lemma 1.11.11. Let Y be a affine irreducible variety, then $Y \times k^r$ is irreducible and

$$\dim Y \times k^r = r + \dim Y \text{ and } \mathcal{O}(Y \times k^r) \cong \mathcal{O}(Y)[T_1, \ldots, T_r]. \tag{1.136}$$

Proof: Let $Y \subset k^s$, then

$$\sum_{\alpha\in\mathbb{Z}^r_{\geq 0}} f_\alpha p_2 T^\alpha \in I(Y \times k^r) \iff \sum_{\alpha\in\mathbb{Z}^r_{\geq 0}} f_\alpha p_2(y) T^\alpha = 0, \ \forall y \in Y. \tag{1.137}$$

So $f_\alpha \in I(Y)$, $\forall \alpha$, therefore $I(Y \times k^r) = p_2^{\#}(I(Y))[T_1, \ldots, T_r]$. Therefore

$$\mathcal{O}(Y \times k^r) = \frac{k[U_1, \ldots, U_s][T_1, \ldots, T_r]}{p_2^{\#}(I(Y))[T_1, \ldots, T_r]} \cong \mathcal{O}(Y)[T_1, \ldots, T_r]. \tag{1.138}$$

However, by Noether normalization lemma, there exits $y_1, \ldots, y_d \in \mathcal{O}(Y)$ such that $\mathcal{O}(Y)$ is integral over $k[y_1, \ldots, y_d]$ where $d = \dim Y$. Therefore $\mathcal{O}(Y \times k^r)$ is integral over $k[y_1, \ldots, y_d][T_1, \ldots, T_r]$. Since $y_1, \ldots, y_d, T_1, \ldots, T_r$ are algebraically independent we conclude that $\dim(Y \times k^r) = d + r$. □

Remark 1.11.12. Let $S \subset R$ be a subring, we say that R is finite over S if there exists $x_1, \ldots, x_m \in R$ such that $R = Sx_1 + \cdots + Sx_m$, i.e., R is a finitely generated $S-$module.

Remark 1.11.13. If X is affine, then any maximal ideal of $\mathcal{O}(X)$ is of the form $I_x = I(\{x\}) = \{f \in \mathcal{O}(X) : f(x) = 0\}$.

Lemma 1.11.14. Let $f : X \to Y$ be a dominant morphism of affine varieties. Suppose that $\mathcal{O}(X)$ is finite over $\mathcal{O}(Y)$, then f is surjective.

Proof: Let $Y' \subset Y$ be an irreducible component and $f^{-1}(Y') = X_1 \cup \cdots \cup X_m$ be irreducible components. Suppose that for all j, there exists $U_j \subset Y'$ such that $U_j \subset Y' - f(X_j)$. Then

$$U = \bigcap_{j=1}^m U_j \subset \bigcap_{j=1}^m (Y' - f(X_j)) = Y' - \bigcup_{j=1}^m f(X_j). \tag{1.139}$$

However, since f is dominant, there exists $f(x) \in f(X) \cap U \Rightarrow x \in f^{-1}(U) \subset f^{-1}(Y')$, so $x \in X_j$ for some j. This contradicts the inclusion $U \subset Y' - \bigcup_{j=1}^m f(X_j)$. this proves that there exists a irreducible components $X' \subset f^{-1}(Y')$ such that $f|_{X'} : X' \to Y'$ is dominant. If we prove that $X' \to Y'$ is surjective, then we are done because we can repeat the same argument for any irreducible component Y' of Y. Furthermore, the hypothesis of the problem transfer to $X' \to Y'$, in fact:

$$\begin{array}{ccc} \mathcal{O}(X) & \twoheadrightarrow & \mathcal{O}(X') \\ \uparrow & & \uparrow \\ \mathcal{O}(Y) & \twoheadrightarrow & \mathcal{O}(Y') \end{array}$$

Remember that since $X' \hookrightarrow X$ is closed, we have $\mathcal{O}(X') = \mathcal{O}(X)/I(X')$ and $\mathcal{O}(X) \to \mathcal{O}(X')$ is the projection, thus it's surjective. Then, we use the above diagram to conclude that $\mathcal{O}(X')$ is finite over $\mathcal{O}(Y')$. Now we can assume both X and Y irreducible. Let $y \in Y$, then we have the diagram

$$\begin{array}{ccccc} \mathcal{O}(X) & \hookrightarrow & \mathcal{O}(X)\mathcal{O}_{Y,y} & \hookrightarrow & K(X) \\ \text{finite}\uparrow & & \text{finite}\uparrow & & \uparrow \\ \mathcal{O}(Y) & \hookrightarrow & \mathcal{O}_{Y,y} & \hookrightarrow & K(Y) \end{array}$$

where

$$\mathcal{O}_{Y,y}\mathcal{O}(X) = \{g/h \circ f : g \in \mathcal{O}(X), h \in \mathcal{O}(Y), h(y) \neq 0\}. \tag{1.140}$$

Now we claim that there exists a maximal ideal $M \subset \mathcal{O}(X)\mathcal{O}_{Y,y}$ such that $M \cap \mathcal{O}_{Y,y} = \mathfrak{m}_{Y,y}$. If this is true, then

$$(M \cap \mathcal{O}(X)) \cap \mathcal{O}(Y) = M\mathcal{O}(Y) = (M \cap \mathcal{O}_{Y,y}) \cap \mathcal{O}(Y) = \mathfrak{m}_{Y,y}\mathcal{O}(Y) = I_y. \tag{1.141}$$

Therefore, for any maximal ideal $I_x \supset M \cap \mathcal{O}(X)$, we have $I_x \supset f^{\#}(I_y) = I_{f(y)}$, which implies that $x = f(y)$.

Write $R = \mathcal{O}(X)\mathcal{O}_{Y,y}$ and $S = \mathcal{O}_{Y,y}$ and let $M \subset R$ be a maximal ideal. We claim that $M \cap S = \mathfrak{m}_{Y,y}$. To show this, it's sufficient to prove that $M \cap S \subset S$ is maximal (because S is local), so we need to prove that $S/(M \cap S)$ is a field. However $S/(M \cap S) \hookrightarrow R/M$ while R is integral over S. So we need to show that if a ring A embedds in a field k, which is integral over A, then A is a field. This is an easy exercise. □

Theorem 1.11.15. Let $f : X \to Y$ be a dominant morphism of irreducible varieties, X quasicompact. Then there exists $U \subset Y$ open dense such that

(1) $U \subset f(X)$;

(2) $\forall\, W \subset Y$ irreducible closed such that $W \cap U \neq \varnothing$ and any irreducible component $Z \subset f^{-1}(W)$ such that $Z \cap f^{-1}(U) \neq \varnothing$, we have

$$\operatorname{codim}(Z, X) = \operatorname{codim}(W, Y). \tag{1.142}$$

Proof:TODO:

Corollary 1.11.16. Let $f : X \to Y$ be a dominant morphism of irreducible varieties, X quasicompact. Then, there exists $U \subset f(X)$ dense, such that $\forall\ y \in U$, $f^{-1}(y)$ is nonempty of pure dimension equal to $r = \dim(X) - \dim(Y)$.

1.11.2 Constructible subsets

Definition 1.11.17. Let X be a topological space. Then we define:

(1) A subset of the form $U \cap F \subset X$, where $U \subset X$ is open and $F \subset X$ is closed is said to be locally closed.

(2) A constructible subset $E \subset X$ is a finite union of locally closed subsets in X.

Remark 1.11.18. From definion above, open and closed sets are constructible.

Proposition 1.11.19. Let $f : X \to Y$ be a morphism of quasicompact algebraic varieties. Then f sends constructible subsets to constructible subsets.

Proof: By definition it's sufficient to prove that the image of a locally closed subset is constructible. However if $E \subset X$ is locally closed, it has a structure of an algebraic variety such that the inclusion $E \hookrightarrow X$ is a morphism. In particular $f|_E : E \to Y$ is a morphism, so we can take $E = X$. Now we are reduced to show that $f(X)$ is constructible in Y. Furthermore, if $X = \bigcup_{i=1}^m X_i$ are the irreducible components of X, then $f(X) = \bigcup_{i=1}^m f(X_i)$, so we are reduced to prove that $f(X_i)$ is constructible for each i. Then we assume X irreducible. A final reduction, let $f(X) \subset Z \subset Y$, where Z is closed in Y. Then $f(X)$ constructible in Z means that

$$f(X) = \bigcup_i (U_i \cap Z) \cap (F_i \cap Z) = \bigcup_i U_i \cap (F_i \cap Z) \tag{1.143}$$

where $U_i \subset Y$ is open and $F_i \subset Y$ is closed for each i. But $U_i \cap (F_i \cap Z)$ is locally closed in Y, so $f(X)$ is constructible in Y. Then we can assume $Y = \overline{f(X)}$, so f is a dominant morphism of quasicompact, irreducible algebraic varieties. Now we are in position to apply the previous theorem to conclude that there exists an open dense $U \subset f(X)$. Now, let $Y - U = \bigcup_{i=1}^m Z_i$ be a decomposition into irreducible components and let $f^{-1}(Z_i) = W_{i1} \cup \cdots \cup W_{it_i},\ i = 1, \ldots, m$ be a decomposition into irreducible components. Then $g_{ij} = f|_{W_{ij}} : W_{ij} \to Z_i$ is a morphism for each $i = 1, \ldots, m$ and $j,\ 1 \leq j \leq t_i$. Furthermore $\dim Z_i < \dim Y$, so we can apply induction on dimenion of Y to conclude that each $f(W_{ij}) \subset Z_i$ is constructible, thus constructible in Y. However

$$f(X) = U \cup \bigcup_{i,j} f(W_{i,j}) \tag{1.144}$$

is constructible. □

Remark 1.11.20. The base case in the previous definition is $\dim Y = 0$, i.e., Y is a point. In this case the result is clearly true.

1.12 Lecture 19 and 20

From now on, we will assume most of the time that X is quasicompact.

1.12.1 Cartier divisors

Definition 1.12.1. A rational function $f : X \dashrightarrow Y$ is an equivalence class of pairs (U, f), $f \in \mathcal{O}_X(U)$ with $U \subset X$ open dense, where the equivalence relation is given by

$$(U, f) \sim (U', f') \iff f|_{U\cap U'} = f'|_{U\cap U'}. \tag{1.145}$$

Notation 1.12.2. To be precise, we will write $\langle U, f\rangle$ to a rational function $f : X \dashrightarrow k$, so it's implicity here that $U \subset X$ is open dense and $f \in \mathcal{O}_X(U)$.

Notation 1.12.3. By definition, a rational function $f : X \dashrightarrow k$ is an element of $K(X)$.

Definition 1.12.4. A cartier divisor on X is a global section $D \in \Gamma(X, K_X^*/\mathcal{O}_X^*)$.

Remember that if $\mathcal{F}$ is a presheaf, then it's sheafification is a sheaf $\mathcal{F}^+$ with sections defined by

$$\mathcal{F}^+(U) = \left\{ s : U \to \bigcup_{P\in U} \mathcal{F}_P \;\middle|\; \begin{array}{c}\text{For each } P \in U, \text{ there exists}\\ \text{a section } t \in \mathcal{F}(V), P \in V \subset U\\ \text{such that } s(Q) = t_Q,\ \forall Q \in V\end{array} \right\} \tag{1.146}$$

It's a sheaf with natural map $\theta : \mathcal{F} \to \mathcal{F}^+$. Its satisfies the following universal property, if $\mathcal{G}$ is a sheaf and $\varphi : \mathcal{F} \to \mathcal{G}$ is a presheaf morphism, then there exists a unique shef morphism $\tilde{\varphi} : \mathcal{F}^+ \to \mathcal{G}$ such that $\tilde{\varphi}\theta = \varphi$. The map is the following, let $s \in \mathcal{F}^+(U)$ and $U = \bigcup_i U_i$ a covering with sections $t_i \in \mathcal{F}(U_i)$ such that $s_P = t_{i,P}$, $\forall P \in U_i$. Then $\varphi(t_i) \in \mathcal{G}(U_i)$ is such that

$$\varphi(t_i)_P = \varphi_P(t_{i,P}) = \varphi_P(s(P)) = \varphi_P(t_{j,P}) = \varphi(t_j)_P \tag{1.147}$$

for all $P \in U_i \cap U_j$. Then, since $\mathcal{G}$ is a sheaf, we have $\varphi(t_i)|_{U_i\cap U_j} = \varphi(t_j)|_{U_i\cap U_j}$, $\forall i, j$. We define $\tilde{\varphi}(s)$ by glueing these sections.

Remark 1.12.5. A section $s \in \mathcal{F}^+(U)$ defines and it's defined by a covering $U = \bigcup_i U_i$ and a collection sections $s_i \in \mathcal{F}(U_i)$ such that $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$. Therefore, doing operations with section s of the sheafification is the same thing as doing operations with the collection of sections $\{s_i\}$ determined s, on a common refinement of coverings.

So, a divisor $D \in \Gamma(X, K_X^*/\mathcal{O}_X^*)$ is defined by a covering $X = \bigcup_i U_i$ and a collection of sections $s_i = [\langle V_i, f_i\rangle] \in P(K_X^*/\mathcal{O}_X^*)(U) = K_X^*(U)/\mathcal{O}_X^*(U)$ such that $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$, which means that

$$\begin{aligned}\langle V_i, f_i\rangle\langle V_j, f_j\rangle^{-1} &= \langle V_i, f_i\rangle\langle\{f_j \neq 0\}, 1/f_j\rangle \\ &= \langle V_i \cap \{f_j \neq 0\}, f_i|_{V_i\cap\{f_j\neq 0\}}/f_j|_{V_i\cap\{f_j\neq 0\}}\rangle \in \mathcal{O}_X^*(U_i \cap U_j).\end{aligned} \tag{1.148}$$

Since the inclusion $1 \to \mathcal{O}_X^* \hookrightarrow K_X^*$ is given by $\mathcal{O}_X^*(U) \ni f \mapsto \langle U, f\rangle$, this means that there exists $g_{ij} \in \mathcal{O}_X^*(U_i \cap U_j)$ such that

$$\begin{aligned}&f_i/f_j = g_{ij} \text{ on } V_i \cap \{f_j \neq 0\} \cap U_i \cap U_j \iff \\ &f_i = g_{ij} f_j \text{ on } V_i \cap V_j.\end{aligned} \tag{1.149}$$

Corollary 1.12.6. By definitin, a Cartier divisor is given by a covering $X = \bigcup_i U_i$ and a collection of rational maps $\langle V_i, f_i\rangle \in K^*(U_i)$ such that for each i, j, there exists $g_{ij} \in \mathcal{O}^*(U_i \cap U_j)$ such that $f_i = g_{ij} f_j$ on $V_i \cap V_j$. Most of the time we will write a Cartier divisor as a covering $U = \bigcup_i U_i$ and pairs $(U_i, f_i)_{i\in I}$ where $f_i \in K^*(U_i) \cong K^*(X)$.

Notation 1.12.7. We write $\mathrm{CaDiv}(X) = \Gamma(X, K_X^*/\mathcal{O}_X^*)$ for the group of Cartier divisors of X.

Now, we see that a Cartier divisor D defines a covering $X = \bigcup_i U_i$ and a collection of regular maps $g_{ij} \in \mathcal{O}_X^*(U_{ij})$. We claim that $g_{ij}g_{jk}g_{ki} = 1$ on U_{ijk} and $g_{ii} = 1$ on U_i. In fact, on $\{f_j f_k f_i \neq 0\} \overset{\text{open}}{\subset} V_{ijk}$ we have

$$g_{ij}g_{jk}g_{ki} = \frac{f_i}{f_j}\frac{f_j}{f_k}\frac{f_k}{f_i} = 1. \tag{1.150}$$

But $\{f_j f_k f_i \neq 0\} \subset U_{ijk}$ is an open dense subset, so $g_{ij}g_{jk}g_{ki} = 1$. In the same way we verify that $g_{ii} = 1$ on U_i. So the collection $\{g_{ij}\}$, $g_{ij} \in \mathcal{O}_X^*(U_{ij})$ defines a line bundle on X that we call $L(D)$.

Proposition 1.12.8. The assigment $D \mapsto L(D)$ is a surjective group homomorphism $\mathrm{CaDiv}(X) \to \mathrm{Pic}(X)$.

Proof: Let $D = (U_i, f_i)_{i\in I}, E = (V_j, g_j)_{j\in J}$ be two Cartier divisors on X, then

$$D + E = (U_i \cap V_j, f_i g_j)_{(i,j)\in I\times J}. \tag{1.151}$$

Therefore $L(D + E)$ is the line bundle defined by transition maps

$$A_{(i,j),(p,q)} = \frac{f_i g_j}{f_p g_q} = \frac{f_i}{f_p}\frac{g_j}{g_q} \tag{1.152}$$

on $U_{ip} \cap V_{jq}$. But this is the product of the transition functions of $L(D)$ and $L(E)$.

To prove surjectivity, we assume first that X is irreducible. Let $L \in \mathrm{Pic}(X)$ be a line bundle defined by a covering $U = \bigcup_{i\in I} U_i$ and transition maps $g_{ij} \in \mathcal{O}_X^*(U_{ij})$. Now, fix $i \in I$ and define $f_j = g_{ij} : U_j \dashrightarrow k$ (it's a rational function because $U_{ij} \subset U_j$ is open dense). Then the collection $(U_i, f_i)_{i\in I}$ defines a Cartier divisor such that $L(D) = L$. To generalize for non irreducible varieties, we take a line bundle L with local trivializations $\varphi_i : U_i \to k \times U_i$ and add one more trivialization $\varphi : V \to k \times V$ in the following way. Let $X = \bigcup_{j=1}^m X_j$ decomposition into irreducible components, then for each j there exists $\alpha(j) \in I$ such that $U_{\alpha(j)} \cap X_j \neq \varnothing$. Then we define $V = \bigsqcup_{j=1}^m V_j$, where

$$V_j = U_{\alpha(j)} \cap (X - \bigcup_{k\neq j} X_k) \overset{\text{open dense}}{\subset} X_j. \tag{1.153}$$

Then we glue $\varphi_{\alpha(j)}$ to define φ, a local trivialization. Then the argument follows from the irreducible by choosing the dense open set V. □

Proposition 1.12.9. The map $\mathrm{CaDiv}(X) \to \mathrm{Pic}(X)$ has kernel given by the principal divisors $(X, f),\ f \in K(X)$.

Proof: If $D = (U_i, f_i)_{i\in I}$ defines a trivial line bundle, then there exists $h_i \in \mathcal{O}_X^*(U_i)$ such that $h_i^{-1} g_{ij} h_j = 1$. Therefore, on $V_i \cap V_j$ (where $f_i \in \mathcal{O}(V_i),\ V_i \supset U_i$ open dense) we have

$$\frac{f_i}{h_i} = \frac{f_j}{h_j}. \tag{1.154}$$

Then $f_i/h_i \in K(U_i)$ and $f_i/h_i|_{U_i\cap U_j} = f_j/h_j|_{U_i\cap U_j}$. So these functions glue to a rational function $f \in K(X)$. Since $f_i = f h_i,\ h_i \in \mathcal{O}_X^*(U_i)$, it's clear that D and (X, f) defines the same divisor. □

1.12.2 Weil divisors

Notation 1.12.10. Suppose that we have the following situation $Y \subset X$ a closed subset, where X is affine. We will write

$$I_{Y|X} = \{f \in \mathcal{O}(X) : f|_Y = 0\}. \tag{1.155}$$

Of course, we have $V(I_{Y|X}) = Y$.

Let X be a irreducible variety. Remember that the natural map $\mathcal{O}(X) \hookrightarrow \mathcal{O}_{X,Y} : f \mapsto \langle X, f\rangle$ is injective. In fact, if $\langle X, f\rangle = 0$, then there exists $U \subset X$ open, with $U \cap Y \neq \varnothing$, such that $f|_U = 0$. But $U \subset X$ is dense (because X is irreducible), so $f = 0$. Now, we claim that

$$\mathrm{Frac}\ \mathcal{O}_{X,Y} = K(X). \tag{1.156}$$

In fact, let $U \subset X$ be an open affine such that $U \cap Y \neq \varnothing$. Then

$$\mathcal{O}(U) \hookrightarrow \mathcal{O}_{U,Y\cap U} \hookrightarrow K(U). \tag{1.157}$$

But since, U is affine we have an isomorphism $\text{Frac } \mathcal{O}(U) \to K(U) : f/g \mapsto \langle U_g, f/g\rangle$. Therefore, we have a commutative diagram

$$\begin{array}{ccccc} \text{Frac } \mathcal{O}(U) & \hookrightarrow & \text{Frac } \mathcal{O}_{U,U\cap Y} & \hookrightarrow & K(U) \\ & \searrow & & \nearrow & \\ & & \cong & & \end{array}$$

In fact, on the first row of the above diagram we have the chain of maps

$$\frac{f}{g} \mapsto \frac{\langle X, f\rangle}{\langle X, g\rangle} \mapsto \langle X, f\rangle\langle X, g\rangle^{-1} = \langle X, f\rangle\langle U_g, 1/g\rangle = \langle U_g, f/g\rangle. \tag{1.158}$$

This proves that the embedding $\mathcal{O}_{U,U\cap Y} \hookrightarrow K(U)$ is an isomorphism. Finally we have

$$\text{Frac } \mathcal{O}_{X,Y} \cong \text{Frac } \mathcal{O}_{U,U\cap Y} \cong K(U) \cong K(X). \tag{1.159}$$

The first isomorphism is induced by $\mathcal{O}_{X,Y} \to \mathcal{O}_{U,U\cap Y} : \langle V, f\rangle \mapsto \langle V \cap U, f|_{V\cap U}\rangle$. Observe that since Y is irreducible, we have $(U \cap Y) \cap (V \cap Y) \neq \varnothing \Rightarrow U \cap V \neq \varnothing$.

Definition 1.12.11. The group of Weil divisors $\text{Div}(X)$ is the free abelian group generated by the set of closed irreducible subvarieties $Y \subset X$ of codimension one. We write

$$D = \sum_Y n_Y[Y],\ n_Y \in \mathbb{Z} \tag{1.160}$$

an element $D \in \text{Div}(X)$.

Proposition 1.12.12. There exists a homomorphism

$$\text{CaDiv}(X) \to \text{Div}(X) : D \mapsto [D] = \sum_{\text{codim}(Y,X)=1} \text{ord}_Y(D)[Y] \tag{1.161}$$

where the order $\text{ord}_Y(D)$ is defined in the following way. Suppose that D is represented by a covering $X = \bigcup_i U_i$ with rational maps $f_i \in K^*(U_i)$. Let $i \in I$ be such that $U_i \cap Y \neq \varnothing$ and write $f_i = a/b$, $a, b \in \mathcal{O}_{X,Y}$ (we used $K(U_i) \cong K(X)$ here). Then we define

$$\text{ord}_Y(D) = \ell(\frac{\mathcal{O}_{X,Y}}{(a)}) - \ell(\frac{\mathcal{O}_{X,Y}}{(b)}). \tag{1.162}$$

Remark 1.12.13. Will observe that we can take the above covering $U = \bigcup_i U_i$ to be a cover of open affines. In this case, if $Y \cap U_i \neq \varnothing$ we have $\text{Frac } \mathcal{O}(U_i) = \text{Frac } \mathcal{O}_{U_i,U_i\cap Y} = K(U_i)$, so we can take $f_i = a_i/b_i$ with $a_i, b_i \in \mathcal{O}(U_i)$.

The purpose of this section is to prove that the above definition is independent of the various choices made to define $\mathrm{ord}_Y(D)$.

Lemma 1.12.14. Lt X be irreducible and $Y \subset X$ be an irreducible variety of codimension 1. Let $f \in \mathcal{O}_{X,Y} \setminus 0$, then there exists $n \geq 0$ such that $\mathfrak{m}_{X,Y}^n \subset f\mathcal{O}_{X,Y}$.

Proof: Let $U \subset X$ be an open affine such that $U \cap Y \neq \varnothing$ and let $f \in \mathcal{O}_{U,U\cap Y}$. Then $\mathcal{O}_{X,Y} \cong \mathcal{O}_{U,U\cap Y}$. So, if we show that $\mathfrak{m}_{U,U\cap Y}^n \subset f\mathcal{O}_{U,U\cap Y}$, we get $\mathfrak{m}_{X,Y}^n \subset f\mathcal{O}_{X,Y}$. So we can assume X affine. Now, write $I = I_{Y|X} \subset \mathcal{O}(X)$ and let $f \in \mathcal{O}(X) \setminus 0$. If $f \notin I$, then $\langle X, f\rangle \in \mathcal{O}_{X,Y}^*$ and $\langle X, f\rangle\mathcal{O}_{X,Y} = \mathcal{O}_{X,Y}$ so $n = 0$ works. If $f \in I$, then $Y \subset V(f)$ and since $Y \subset X$ is irreducible of codimension 1, it's irreducible maximal in $V(f)$, i.e., it's a irreducible component. Now, there exists an open affine $U \subset X$ such that $U \cap Y = U \cap V(f) = V(f|_U)$ (see proof of theorem (1.10.8)). Then we do a further reduction, we assume $Y = V(f), f \in \mathcal{O}(X)$. By Nullstelensatz $I = \sqrt{(f)}$, which is finitely generated by Hilbert basis, so $I^d \subset (f)$ for some $d \geq 0$. We have

$$\mathcal{O}(X) \to \mathcal{O}_{X,Y} : I \mapsto I\mathcal{O}_{X,Y} = \mathfrak{m}_{X,Y} \tag{1.163}$$

and $\mathfrak{m}_{X,Y}^d \subset \langle X, f\rangle\mathcal{O}_{X,Y}$. □

Proposition 1.12.15. Let A be a local Noetherian ring and $P \subset A$ a maximal ideal, M a finitely generated $A-$modulo such that $P^n M = 0$ for some $n \geq 0$. Then the series

$$0 = P^n M \subset P^{n-1}M \subset \cdots \subset PM \subset M \tag{1.164}$$

can be refined to a composition series of length

$$\ell_A(M) = \sum_{i=1}^{n} \dim_{A/P}\left(\frac{P^{i-1}M}{P^i M}\right). \tag{1.165}$$

As a consequence, we have that if $Y \subset X$ is irreducible of codimension one, and $f \in \mathcal{O}_{X,Y} \setminus 0$, then $\ell(\frac{\mathcal{O}_{X,Y}}{(f)}) < \infty$.

Proposition 1.12.16. Let $Y \subset X$ be an irreducible variety of codimension one, then

$$\ell\left(\frac{\mathcal{O}_{X,Y}}{(fg)}\right) = \ell\left(\frac{\mathcal{O}_{X,Y}}{(f)}\right) + \ell\left(\frac{\mathcal{O}_{X,Y}}{(g)}\right). \tag{1.166}$$

Proof: This follows from the exact sequence

$$0 \to \frac{\mathcal{O}_{X,Y}}{(g)} \xrightarrow{\times f} \frac{\mathcal{O}_{X,Y}}{(fg)} \to \frac{\mathcal{O}_{X,Y}}{(f)} \to 0. \tag{1.167}$$

□

Proof of proposition (1.12.12): Let $Y \subset X$ be irreducible of codimension 1 and let $i \in I$ be such that $U_i \cap Y \neq \varnothing$, also let $f_i = a/b$ with $a, b \in \mathcal{O}(U_i)$ (see the remark above the proposition). Then the equality

$$\operatorname{ord}_Y(D) = \ell(\frac{\mathcal{O}_{X,Y}}{a_i}) + \ell(\frac{\mathcal{O}_{X,Y}}{b_i}) \tag{1.168}$$

(1) Does not deppend on (a, b): If $a/b = c/d$, then $ad = bc$ and $\ell(\frac{\mathcal{O}_{X,Y}}{ad}) = \ell(\frac{\mathcal{O}_{X,Y}}{bc})$ and we are done because of the previous proposition.

(2) Does not deppend on $i \in I$: Let $j \in J$ be such that $U_j \cap Y \neq \varnothing$, so $U_i \cap U_j \cap Y \neq \varnothing$, then $f_j = g_{ji} f_i$, with $g_{ji} \in \mathcal{O}_X^*(U_i \cap U_j)$. Then $f_j = \frac{a g_{ji}}{b}, g_{ji} \in \mathcal{O}_{X,Y}^*$, so $\ell(\frac{\mathcal{O}_{X,Y}}{(g_{ji}a)}) = \ell(\frac{\mathcal{O}_{X,Y}}{(a)})$.

(3) Does not deppend on the representative of the divisor: Let $D = (V_l, h_l)_{l \in L}$ and $V_l \cap Y \neq \varnothing$. Then $h_l = f_i g, g \in \mathcal{O}_X^*(V_l \cap U_i)$ and the result follows like in the previous item.

(4) We have $\operatorname{ord}_Y(D + D') = \operatorname{ord}_Y(D) + \operatorname{ord}_Y(D')$, because of the previous proposition and the definition of the sum of two divisors.

(5) The sum $\sum_Y \operatorname{ord}_Y(D)[Y]$ is finite. Since X is quasicompact, we can take $D = (U_i, f_i)_{i \in I}$, $|I| < \infty$ and U_i affine. So $f_i = a_i/b_i$, $a_i, b_i \in \mathcal{O}_X(U_i)$. Let $Y \subset X$ be of codimension one and $U_i \cap Y \neq \varnothing$. We knwo that both $V(a_i)$ and $V(b_i)$ has a finite number of irreducible components of codimenion one in U_i. Then, if $U_i \cap Y \subset V(a_i) \cup V(b_i)$, it must be in this (finite) set of irreducible components, because $\operatorname{codim}(U_i \cap Y, U_i) = \operatorname{codim}(Y, X) = 1$. So we claim that if $\operatorname{ord}_Y(D) \neq 0$, then $U_i \cap Y$ is an element of this collection. If this is not the case, then $a_i|_{U_i \cap Y}, b_i|_{U_i \cap Y} \not\equiv 0$, which means that $\langle U_i, a_i \rangle, \langle U_i, b_i \rangle \in \mathcal{O}^*_{U_i, U_i \cap Y} = \mathcal{O}^*_{X,Y}$, so

$$\ell(\frac{\mathcal{O}_{X,Y}}{a_i}) = \ell(\frac{\mathcal{O}_{X,Y}}{b_i}) = 0. \tag{1.169}$$

Definition 1.12.17. We define a linear equivalence relation on $\operatorname{CaDiv}(X)$ by $D \sim D' \iff D - D' = (X, f)$ for some $f \in \mathcal{O}(X)$ and in a similar way for $\operatorname{Div}(X)$. We denote the respective quotient groups by $\operatorname{CaCl}(X)$ and $\operatorname{Cl}(X)$.

Notation 1.12.18. Normally, we will write $\operatorname{div}(f) = [(X, f)] \in \operatorname{Div}(X)$.

Example 1.12.19. Let X be an affine variety such that $\mathcal{O}(X)$ is an UFD. Then $\operatorname{Cl}(X) = 0$, in fact if $Y \subset X$ is irreducible of codimention 1 and $f \in I = I_{Y|X} \setminus 0$ is irreducible (every prime ideal in a UFD contains an irreducible element), then $Y \subset V(f)$, so $Y = V(f)$ because they both have the same dimension. Now, if $Z \subset X$ has codimension 1, then $Z \neq Y \Rightarrow f|_Z \not\equiv 0$ and $f \in \mathcal{O}^*_{X,Z}$. This implies

$$\operatorname{ord}_Z(X, f) = \ell(\frac{\mathcal{O}_{X,Z}}{(f)}) = 0. \tag{1.170}$$

So $\operatorname{div}(f) = [(X, f)] = [Y]$.

Lemma 1.12.20. Let X be an irreducible variety, $Y \subset X$ a proper irreducible closed subset of codimension 1. If $U = X - Y$, then we have a surjection $\operatorname{Cl}(X) \to \operatorname{Cl}(U)$ whose kernel is generated by $[Y]$.

Proof: We claim that the map defined by

$$|_U : \operatorname{Div}(X) \to \operatorname{Div}(U) : \begin{cases} Z \mapsto [U \cap Z], \text{ if } Z \cap U \neq \varnothing \\ 0 \text{ otherwise} \end{cases} \tag{1.171}$$

is a surjection. In fact, if $Z \subset U$ is a closed irreducible subvariety of codimension 1, then $\overline{Z} \subset X$ is irreducible of codimension 1. This follows because Z is irreducible in its subspace topology if and only if $\overline{Z} \subset X$ is irreducible. Furthermore since $Z \subset U$ is closed, then $Z = U \cap F$ where $F \subset X$ is closed. Therefore $\overline{Z} \cap U \subset \overline{U} \cap F \cap U = F \cap U \Rightarrow \overline{Z} \cap U = Z$. Therefore $Z \subset \overline{Z}$ is open and $\dim \overline{Z} = \dim Z$, so $\operatorname{codim}(Z, U) = \operatorname{codim}(Z, X)$. This proves that there exists a bijection of irreducible codimension 1 subvarieties of U and irreducible codimension 1 subvarieties of X intersecting U. This proves that $|_U : \operatorname{Div}(X) \to \operatorname{Div}(U)$ is surjective. Now we need to prove that its induces a well defined map on quotients. Let $\operatorname{div}(f) = \sum_Z \operatorname{ord}_Z(X, f)[Z]$, we claim that $\operatorname{div}(f)|_U = \operatorname{div}(f|_U)$. This will follows if we show that

$$\operatorname{ord}_Z(X, f) = \operatorname{ord}_Z(U, f|_U) \tag{1.172}$$

whenever $U \cap Z \neq \varnothing$. However, we have

$$\operatorname{ord}_Z(X, f) = \ell(\frac{\mathcal{O}_{X,Z}}{\langle V, f\rangle}) = \ell(\frac{\mathcal{O}_{U,U\cap Z}}{\langle V \cap U, f|_{V\cap U}\rangle}) = \operatorname{ord}_Z(U, f|_U) \tag{1.173}$$

because of the isomorphism $\mathcal{O}_{X,Z} \to \mathcal{O}_{U,U\cap Z} : \langle V, g\rangle \mapsto \langle V \cap U, g|_{V\cap U}\rangle$. Now suppose that $D = \sum_Z n_Z[Z]$ is mapped to zero. Then there exists $f \in K(U) \cong K(X)$ such that $D|_U = \sum_{Z\cap U\neq\varnothing} n_Z[Z \cap U] = \operatorname{div}(f|_U)$. Therefore, we have

$$D - \operatorname{div}(f) = D - \sum_{Z\cap U\neq\varnothing} n_Z[Z] = n_Y[Y] \tag{1.174}$$

which shows that $D \sim n_Y[Y]$. □

Corollary 1.12.21. With the conditions of the previous lemma, we have an exact sequence

$$\mathbb{Z} \to \operatorname{Cl}(X) \to \operatorname{Cl}(U) \to 0. \tag{1.175}$$

Remark 1.12.22. In the previous lemma, if Y has codimension ≥ 2, we can show in the same lines that $\operatorname{Cl}(X) \cong \operatorname{Cl}(U)$.

Example 1.12.23. Let $X = V(UV - W^2) \subset k^3$ and $Y = V(U, W) \subset X$. We claim that $\mathrm{Cl}(X) \cong \mathbb{Z}/2$. We have $D_X(U) \subset X \supset X - D_X(U) = Y$ and

$$\mathcal{O}_X(D_X(U)) = (\frac{k[U,V,W]}{(UV - W^2)})_U \cong \frac{k[U,V,W][1/U]}{(UV - W^2)} \cong k[U, U^{-1}, W] \tag{1.176}$$

which is a UFD, therefore $\mathrm{Cl}(D_X(U)) = 0$ by example (1.12.19). Now, by the previous lemma we have an exact sequence

$$\mathbb{Z} \to \mathrm{Cl}(X) \to \mathrm{Cl}(D_X(U)) \to 0 \tag{1.177}$$

and since $\mathrm{Cl}(D_X(U)) = 0$, the map $\mathbb{Z} \to \mathrm{Cl}(X)$ is surjective, i.e., $\mathrm{Cl}(X) = \mathbb{Z}[Y]$. Now we claim that $\mathrm{div}(U) = 2[Y] \sim 0$. Rememeber $\mathrm{ord}_Z(X, U) = \ell(\frac{\mathcal{O}_{X,Z}}{\langle X, U\rangle})$. Also, $U|_Z = 0 \to U \in I_{Z|X}$ and $Z \subset V_X(U) = Y \Rightarrow Z = Y$ because both have codimension one in X. Therefore $Z \neq Y$ implies that $U|_Z \neq 0$ and $\langle X, U\rangle \in \mathcal{O}^*_{X,Z}$, so $\mathrm{ord}_Z(X, U) = 0$. Therefore $\mathrm{div}(U) \in \mathbb{Z}[Y]$ (actually, since we already have proved that $\mathrm{Cl}(X) = \mathbb{Z}[Y]$, this passage looks unnecessary). Now, since $\mathcal{O}_{X,Y} \cong \mathcal{O}_{D_X(V),Y\cap D_X(V)}$, then $\ell(\frac{\mathcal{O}_{X,Y}}{\langle X,U\rangle)}) = \ell(\frac{\mathcal{O}_{D_X(V),Y\cap D_X(V)}}{\langle D_X(V),U\rangle})$. But

$$(W) = I_{Y\cap D_X(V)|D_X(V)} \subset \mathcal{O}_X(D_X(V)) \cong k[V, V^{-1}, W]. \tag{1.178}$$

The first equality is a consequence of $Y \cap D_X(V) = V(W) \cap D_X(V)$. Therefore we see that $\mathfrak{m}_{D_X(V),Y\cap D_X(V)} = W\mathcal{O}_{D_X(V),Y\cap D_X(V)}$. Now, since $U = W^2/V$ on $\mathcal{O}_{X,Y}$ (because, $V|_Z \not\equiv 0$), we have the following maximal chain of submodules:

$$\begin{array}{ccccc} (0) & \subset & \frac{\mathfrak{m}_{X,Y}}{(U)} & \subset & \frac{\mathcal{O}_{X,Y}}{(U)} \\ & & & & \\ & & 2 & 1 & 0 \end{array}$$

Then we conclude that $\ell(\frac{\mathcal{O}_{X,Y}}{\langle X,U\rangle)}) = 2$, so $\mathrm{div}(U) = 2[Y]$.

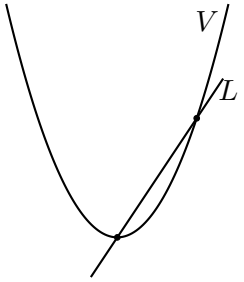
Remark 1.12.24. We have used several times the identity $\mathfrak{m}_{X,Y} = I_{Y|X}\mathcal{O}_{X,Y}$.

1.13 Lecture 21 and 22

Let $V = V(f_1, \ldots, f_m)$ be an affine variety and $x \in V$. Define the line $L = \{x + tv : t \in k\}$ for $v \in k^n$. Then we have the bijection

$$V \cap L \leftrightarrow \{t \in k : f_i(x + tv) = 0,\ \forall i\} \tag{1.179}$$

Since $f_i(x + tv) = t \sum D_i(f)|_x v_i + \mathcal{O}(t^2)$, we see that $t = 0$ is a "multiple root of f" if and only if $\sum_i D_i(f)|_x v_i = 0$.



Here we see a secant line to V. In this case, we have a solution $t \neq 0$ to the set of equations $f_i(x + tv) = 0$. Therefore, $t = 0$ is not a multiple root.

This heuristic introduction motivates the definition for the tangent space at $x \in V$.

Definition 1.13.1. (Tangent space definition 1). We define

$$T_x V = \{v \in k^n : \sum_i D_i(f)|_x v_i = 0, \ \forall f \in I(V)\}. \tag{1.180}$$

Notation 1.13.2. Let M be a $A-$module, where A is a $R-$algebra. Write $\mathrm{Der}_R(A, M)$ to the set of derivations $D : A \to M$. That is, $R-$module morphisms satisfying the Leibniz rule

$$D(ab) = b \cdot Da + a \cdot Db. \tag{1.181}$$

If $v \in T_x V$ and $D = \sum_i v_i D_i \in \mathrm{Der}_k(k[\underline{X}], k[\underline{X}])$. Then we see that $DI \subset \mathfrak{m}_x$, where $I = I(V)$. Then we get an induced map

$$D : k[V] \to k[V]/\mathfrak{m}_x \cong k : f \mapsto D(f)|_x. \tag{1.182}$$

This map is an element of $\mathrm{Der}_k(k[V], k_x)$, where $k = k_x$ is the $k[V]-$module defined by $f \cdot a = f(x)a$ for $f \in k[V]$ and $a \in k$. Also, we see that each element of $\mathrm{Der}_k(k[V], k_x)$ is obtained from a $v \in T_x V$ in this way. In fact, if $D \in \mathrm{Der}_k(k[V], k_x)$, then

$$\begin{aligned} D(\overline{X}^\alpha) &= \sum_i D(\overline{X}_i)\overline{X}^{\alpha - e_i}\big|_x \quad \text{(Leibniz rule)} \\ &= \sum_i v_i D_i(\overline{X}^\alpha)\big|_x. \end{aligned} \tag{1.183}$$

So if $D = \sum_i v_i D_i$. To see that $v = (v_1, \ldots, v_n) \in T_x V$ we calculate

$$\sum_i v_i D_i(f)|_x = D(\overline{f}), \tag{1.184}$$

which is zero whenever $f \in I(V) \iff \overline{f} = 0$. So we get a bijection

$$T_xV \to \mathrm{Der}_k(k[V], k_x) : v \mapsto D_v = \sum_i v_i D_i \tag{1.185}$$

and inverse $D \mapsto (D(\overline{X}_i))_{i=1}^n$. This motivates another definition for the tangent space.

Definition 1.13.3. (Tangent space, definition 2). We define

$$T_xV = \mathrm{Der}_k(k[V], k_x). \tag{1.186}$$

If $D \in \mathrm{Der}_k(k[V], k_x)$, we observe that $D(\mathfrak{m}_x^2) = 0$. In fact, if $f, g \in \mathfrak{m}_x$, then $D(fg) = g(x)D(f) + f(x)D(g) = 0$. So D induces a map $\lambda(D) : \mathfrak{m}_x/\mathfrak{m}_x^2 \to k : f + \mathfrak{m}_x^2 \mapsto D(f)$.

Lemma 1.13.4. The map $\lambda : T_xV \to (\mathfrak{m}_x/\mathfrak{m}_x^2)^*$ is a bijection.

Proof: We prove the lemma by defining a inverse

$$\mu : (\mathfrak{m}_x/\mathfrak{m}_x^2)^* \ni \ell \mapsto [\mu(\ell) : f \mapsto \ell(f - f(x) + \mathfrak{m}_x^2)]. \tag{1.187}$$

To see that $\mu(\ell)$ is a derivation, we compute.

$$\begin{aligned} \mu(\ell)(fg) &= \ell(fg - fg(x) + \mathfrak{m}_x^2) \\ &= \ell(fg - fg(x) - (f - f(x))(g - g(x)) + \mathfrak{m}_x^2) \\ &= \ell((f - f(x))g(x) + (g - g(x))f(x) + \mathfrak{m}_x^2) \\ &= g \cdot \mu(\ell)(g) + f \cdot \mu(\ell)(g). \end{aligned} \tag{1.188}$$

Furthermore, we have

$$\begin{aligned} \mu\lambda(D)(f) &= \lambda(D)(f - f(x) + \mathfrak{m}_x^2) \\ &= D(f - f(x)) \\ &= D(f) \end{aligned} \tag{1.189}$$

and

$$\begin{aligned} \lambda\mu(\ell)(f + \mathfrak{m}_x^2) &= \mu(\ell)(f) \\ &= \ell(f - f(x) + \mathfrak{m}_x^2) \\ &= \ell(f + \mathfrak{m}_x^2) \quad (\text{because } f(x) = 0). \end{aligned} \tag{1.190}$$

□

This motivates the definition of a cotangent space $T_x^*V = \mathfrak{m}_x/\mathfrak{m}_x^2$.

Definition 1.13.5. Let $f = (f_1, \ldots, f_n) : X \subset k^m \to Y \subset k^n$ be a morphism of affine algebraic varieties, we define

$$df_x : T_xX \to T_{f(x)}Y : D \mapsto D \circ f^* \tag{1.191}$$

where we use the second definition of tangent space. At the level of the first definition, the derivative has the form $df_x = J(f)_x$, where $J(f)_x = (D_j(f_i))_{1\leq i\leq n, 1\leq j\leq m}$. This can be seen by

$$v \mapsto D_v \mapsto D_v \circ f^* \mapsto (D_v \circ f^*(\overline{X}_i))_{i=1}^n \tag{1.192}$$

and $f^*(\overline{X}_i) = f_i$, so $(D_v(f_i))_{i=1}^n = J(f)_x \cdot v$.

Recall that if $\phi : A \to B$ is a morphism of $R-$algebras and N is a $A-$module, then B becomes a N-module through ϕ. This is what happens in the previous definition: through f^*, the $k[X]-$module k_x becomes $k_{f(x)}$. In fact, if $g \in k[Y]$ and $a \in k_x$ we have by definition $g \cdot a = f^*(g) \cdot a = f^*(g)|_x a = g(f(x))a$.

Remark 1.13.6. A computational tool: Let $D = k[t]/(t^2)$ be the ring of dual numbers and $\epsilon = t + (t^2)$. Then, if $f \in k[\underline{X}]$, we have $f(x + \epsilon v) = f(x) + \epsilon df_x(v)$, where $df_x(v) = \sum_i D_i(f)|_x v_i$. So, in the context of the previous theorem we can calculate $df_x(v)$ by expanding the $\epsilon-$terms of $f(x+\epsilon v)$. The ring of dual numbers has a multiplicative system given by $S = \{a + \epsilon b : a \neq 0\}$, so we can define its localization $D_S = \{f/g : f \in D,\ g \in S\}$. Let $f, g \in k[X_1 \ldots, X_n]$, then if $x \in D(g)$ we have $g(x + \epsilon v) \in S$ for all $v \in k^n$. So it makes sense to write $\frac{f}{g}(x + \epsilon v) \in D_S$. Moreover, one has

$$\begin{aligned}
\frac{f}{g}(x + \epsilon v) &= \frac{f(x + \epsilon v)}{g(x + \epsilon v)} \\
&= \frac{f(x) + \epsilon df_x(v)}{g(x) + \epsilon dg_x(v)} \\
&= \frac{f(x) + \epsilon df_x(v)}{g(x) + \epsilon dg_x(v)} \frac{g(x) - \epsilon dg_x(v)}{g(x) - \epsilon dg_x(v)} \\
&= \frac{f(x) + \epsilon df_x(v)}{g(x)}(1 - \frac{\epsilon dg_x(v)}{g(x)}) \\
&= \frac{f}{g}(x) + \epsilon d(\frac{f}{g})_x(v).
\end{aligned} \tag{1.193}$$

Now, let $f = (\frac{f_1}{g}, \ldots, \frac{f_n}{g}) : X = D(g) \subset k^m \to k^n$. We see that X is a affine variety contained, but not closed in k^m, so it does not makes sense to talk about the coordinate ring of X. So it still makes sense to talk about the derivative of f in this case ?

Definition 1.13.7. (Tangent space, definition 3). Let $X \subset k^m$ be a locally closed subset, we define

$$T_xX = \{v \in k^m : [\mathcal{O}(X) \ni f \mapsto f(x + \epsilon v) \in D] \in \mathrm{Hom}_{k-\mathrm{alg}}(\mathcal{O}(X), D)\}. \quad (1.194)$$

This definition agrees with the previous ones when $X \subset k^m$ is closed and affine. Furthermore, if $\varphi : X \subset k^m \to Y \subset k^n$ is a morphism of locally closed subsets, we can define the derivative as follows. Let $v \in T_xX$, and define $d\varphi_x(v)$ by the equality $\varphi(x+\epsilon v) = \varphi(x) + \epsilon d\varphi_x(v)$. Then $\mathcal{O}(Y) \ni g \mapsto g(\varphi(x) + \epsilon d\varphi_x(v))$ is a k-algebra morphism. This follows because of the observation: $g(\varphi(x) + \epsilon d\varphi_x(v)) = \varphi^*(g)(x + \epsilon v), \varphi^*(g) \in \mathcal{O}(X)$. Therefore we get a map $d\varphi_x : T_xX \to T_{\varphi(x)}Y : v \mapsto d\varphi_x(v)$.

Example 1.13.8. In the previous example we had $f = (\frac{f_1}{g}, \ldots, \frac{f_n}{g}) : X = D(g) \subset k^m \to k^n$. Then we calculate directly

$$\begin{aligned} f(x + \epsilon v) &= (\frac{f_1}{g}(x + \epsilon v), \ldots, \frac{f_n}{g}(x + \epsilon v)) \\ &= f(x) + \epsilon(d(\frac{f_1}{g})_x(v), \ldots, d(\frac{f_n}{g})_x(v)). \end{aligned} \quad (1.195)$$

Therefore $df_x(v) = (d(\frac{f_1}{g})_x(v), \ldots, d(\frac{f_n}{g})_x(v))$.

Lemma 1.13.9. We have an isomorphism

$$\mathrm{Der}_k(\mathcal{O}_{X,x}, k_x) \to \mathrm{Der}_k(k[X], k_x) : D \mapsto D|_{k[X]}. \quad (1.196)$$

Where we regard $k[X] \subset \mathcal{O}_{X,x}$ through the embedding $f \mapsto \langle X, f\rangle$.

Proof: It's sufficient to give an inverse, to do this we remember that $\mathcal{O}_{X,x} \cong k[X]_{\mathfrak{m}_x}$ (see (1.7.4)). So any derivation $D \in \mathrm{Der}_k(k[X], k_x)$ can be extended to $\mathcal{O}_{X,x}$ using the usual rule $D(f/g) = g(x)^{-2}(g \cdot D(f) - f \cdot D(g))$. Explictly, it has the form

$$D(\langle U, \frac{f}{g}\rangle) = g(x)^{-2}(g \cdot D(f) - f \cdot D(g)). \quad (1.197)$$

□

Lemma 1.13.10. Let $\phi : X \xrightarrow{\cong} U \subset Y$, where U is open affine. Then $d\phi_x : T_xX \to T_{f(x)}Y$ is an isomorphism.

Proof: The isomorphism $\phi^* : k[U] \to k[X]$ induces an isomorphism

$$\mathcal{O}_{U,\phi(x)} \cong k[U]_{\mathfrak{m}_{f(x)}} \to k[X]_{\mathfrak{m}_x} \cong \mathcal{O}_{X,x}, \quad (1.198)$$

because $f \in \mathfrak{m}_{\phi(x)} \iff \phi^*(f) \in \mathfrak{m}_x$. Finally, the isomorphism $\mathcal{O}_{Y,\phi(x)} \cong \mathcal{O}_{U,\phi(x)}$ induces an isomorphism $\mathrm{Der}_k(\mathcal{O}_{X,x}, k_x) \to \mathrm{Der}_k(\mathcal{O}_{Y,\phi_x}, k_{\phi(x)})$. □

Definition 1.13.11. Let X be an abstract variety, we define a directed system (T_xU, ρ_{UV}), $U \ni x$ where ρ_{UV} is the isomorphism $T_xU \to T_xV$ induced by the inclusion $U \subset V$ of open affine subsets. Then we define

$$T_xX = \varinjlim_{U \ni x} T_xU. \tag{1.199}$$

The derivative $d\phi_x$ is defined as follows, let $V \ni \phi(x)$ be an open affine. Then there exists an open affine $U \ni x, U \subset \phi^{-1}(V)$. Then we send

$$d\phi_x : \langle T_xU, D\rangle \mapsto \langle V, d(\phi|_U)_x(D)\rangle. \tag{1.200}$$

The definition of $d\phi_x$ makes sense because

$$\begin{array}{ccc} U & \subset & U' \\ \downarrow{\scriptstyle \phi|_U} & & \downarrow{\scriptstyle \phi|_{U'}} \\ V & \subset & V' \end{array} \rightsquigarrow \begin{array}{ccc} T_xU & \subset & T_xU' \\ \downarrow{\scriptstyle d(\phi|_U)_x} & & \downarrow{\scriptstyle d(\phi|_{U'})} \\ T_xV & \subset & T_xV' \end{array}$$

Chapter 2

Algebraic Groups

2.1 Some notes from Geck

This is some notes that i did based on the first chapter of the book [1]. Just basic/ preliminary stuff.

2.2 Polynomial rings

We define $k[X_1, \ldots, X_n]$ as the monoid algebra defined by kM, where M is the monoid $(\mathbb{Z}_{\geq 0}^n, +)$. For $\alpha \in \mathbb{Z}_{\geq 0}^n$ we write X^α to the associated element in kM through the injection $\mathbb{Z}_{\geq 0}^n \hookrightarrow M$. In particular $X^\alpha \cdot X^\beta = X^{\alpha+\beta}$ by definition. Let $e_i = (\delta_{ji})_{j=1}^n \in M$, then we write $X_i = X^{e_i}$. So we have the formula

$$X^\alpha = X_1^{\alpha_1} \cdots X_n^{\alpha_n}. \tag{2.1}$$

By definition, the collection $\{X^\alpha\}_{\alpha \in M}$ is linearly independent. So, for any $k-$algebra A and elements $a_i \in A,\ i = 1, \ldots, n$ we have an unique linear map $k[X_1, \ldots, X_n] \to A : X^\alpha \mapsto \prod_i a_i^{\alpha_i}$. This linear map is obviously a morphism of $k-$algebras.

In an analogous way, we can define $k[X_1^{\pm}, \ldots, X_n^{\pm}]$ as the group algebra $k\mathbb{Z}^n$. In particular, we can obtain this algebra by doing localizations. For example we have

$$k[X]_X \cong k[X, X^{-1}]. \tag{2.2}$$

2.3 Basic definitions

Some important observations:

- If $\mathbb{F}$ is a finite field, the map which takes a polynomial in $\mathbb{F}[X_1, \ldots, X_n]$ and send to a function $\mathbb{F}^n \to \mathbb{F}$ is not an isomorphism. For example, if $\mathbb{F}$ has q elements, then

$\prod_{q\in\mathbb{F}}(X-q) \in \mathbb{F}[X] \mapsto 0$. However, if $\mathbb{F}$ is infinity and $a_nX^n+a_{n-1}X^{n-1}+\cdots+a_0 = 0$, then we can select distinc elements $\alpha_0, \ldots, \alpha_n$ and we conclude that

$$\begin{pmatrix} a_0 & \ldots & a_n \end{pmatrix} \begin{pmatrix} 1 & 1 & \cdots & 1 \\ \alpha_0 & \alpha_1 & \cdots & \alpha_n \\ \vdots & \vdots & \cdots & \vdots \\ \alpha_0^n & \alpha_1^n & \cdots & \alpha_n^n \end{pmatrix} = 0. \tag{2.3}$$

However $\det(\alpha_i^j)_{i,j=0}^n = \prod_{i<j}(\alpha_i - \alpha_j) \neq 0$ and we conclude that $a_0, \ldots, a_n = 0$. Now, if $f \in \mathbb{F}[X_1, \ldots, X_n]$, then we can write (assumin X_1 appears in f):

$$f = g_d(X_2, \ldots, X_n)X_1^d + \cdots + g_1(X_2, \ldots, X_n)X_1 + g_0(X_2, \ldots, X_n). \tag{2.4}$$

So the equality $f = 0$, the previous step for the case $\mathbb{F}[X]$ and an induction on the number of variables yields $g_d, \ldots, g_0 = 0$. Then we see that if $|\mathbb{F}| = \infty$, we can identify the ring $\mathbb{F}[X_1, \ldots, X_n]$ with polynomial functions $\mathbb{F}^n \to \mathbb{F}$.

As a consequence of this story, we see that if $|\mathbb{F}| = \infty$, then

$$I(\mathbb{F}^n) = (0) \subset \mathbb{F}[X_1, \ldots, X_n]. \tag{2.5}$$

- (Hilbert basis theorem). The Hilbert basis theorem states that if A is noetherian, then $A[X]$ is noetherian. Since $\mathbb{F}[X]$ is a PID (and in consequence noetherian), we conclude by an induction on the number of variables that $\mathbb{F}[X_1, \ldots, X_n]$ is noetherian.

  Remark 2.3.1. A ring is said to be noetherian if it satisfy the ascending chain condition, or equivalently, if all of its ideals are finitely generated.

- (Algebraic sets). Let $S \subset \mathbb{F}[X_1, \ldots, X_n]$, then $V(S) = \{a \in \mathbb{F}^n : f(a) = 0\}$. It's immediate that $V(S) = V((S))$. The relation $V(\bigcup_{i\in I} S_i) = \bigcap_{i\in I} V(S_i)$ allow us to define the Zariski topology on $\mathbb{F}^n$. Moreover, for any subset $V \subset \mathbb{F}^n$, we let $I(V) = \{f \in \mathbb{F}[X_1, \ldots, X_n] : f(a) = 0,\ \forall a \in V\}$. Then we have

$$\begin{cases} VI(W) = \overline{W} \\ IV(\mathfrak{a}) = \sqrt{\mathfrak{a}} \end{cases} \tag{2.6}$$

  We also have the important relation

$$V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b}). \tag{2.7}$$

  In fact

$$V(\mathfrak{a}) \cup V(\mathfrak{b}) \underset{\mathfrak{a}\cap\mathfrak{b}\subset\mathfrak{a},\mathfrak{b}}{\subset} V(\mathfrak{a} \cap \mathfrak{b}) \underset{\mathfrak{a}\mathfrak{b}\subset\mathfrak{a}\cap\mathfrak{b}}{\subset} V(\mathfrak{a}\mathfrak{b}) = V(\{fg : f \in \mathfrak{a}, g \in \mathfrak{b}\}) \subset V(\mathfrak{a}) \cup V(\mathfrak{b}). \tag{2.8}$$

- (Coordinate rings). Let $\mathfrak{a} = I(V)$ where V is an algebraic set. We define the coordinate ring $A[V] = \mathbb{F}[X_1, \ldots, X_n]/\mathfrak{a}$.

- (LEX order). We define an ordering on $\mathbb{Z}^n_{\geq 0}$ by $\alpha > \beta$ if for the smallest i such that $\alpha_i \neq \beta_i$ we have $\alpha_i > \beta_i$. So, with a fixed LEX on $\mathbb{F}[X_1, \ldots, X_n]$, we can define the leading term of a polynomial: $\mathrm{LT}(f) = a_\alpha X^\alpha$, where α is the smallest tuple, relative to LEX such that $a_\alpha \neq 0$. In the case $\mathrm{LT}(f) = a_\alpha X^\alpha$, we also write $\mathrm{LM}(f) = X^\alpha$.

- (Groebner Basis). Let $\mathfrak{a}$ be an ideal in $\mathbb{F}[X_1, \ldots, X_n]$. A groebner basis of $\mathfrak{a}$ is a finite subset $G \subset \mathfrak{a}$ such that $(\mathrm{LT}(g) : g \in G) = \mathrm{LT}(\mathfrak{a})$.

  Example 2.3.2. Let $R = \mathbb{F}[X_1, \ldots, X_n, Y_1, \ldots, Y_m]$ with LEX: $X_1 > \cdots > X_n > Y_1 > \cdots > Y_m$ and define an ideal $\mathfrak{a} = (X_1 - g_1, \ldots, X_n - g_n)$ for $g_i \in \mathbb{F}[Y_1, \ldots, Y_m]$. Since $\mathrm{LT}(X_i - g_i) = X_i$, it's clear that $(X_1, \ldots, X_n) \subset \mathrm{LT}(I)$. Now, if $f \in R$ and $X_j \nmid \mathrm{LT}(f)$, $\forall j$, then $f \in \mathbb{F}[Y_1, \ldots, Y_m]$ because its smallest term has power $\alpha \in \{0\} \times \mathbb{Z}^n$. So, we need to show that $I \cap \mathbb{F}[Y_1, \ldots, Y_m] = (0)$. This can be seen by proving that the kernel of $\varphi : R \to \mathbb{F}[Y] : X_i \mapsto g_i, Y_i \mapsto Y_i$ is I. It's clear that $I \subset \mathrm{Ker}(\varphi)$, so we have an induced morphism $\overline{\varphi} : R/I \to \mathbb{F}[\underline{Y}]$. However, it has an inverse given by $q : \mathbb{F}[\underline{Y}] \to R/I : Y \mapsto [Y]$. Therefore, if $\varphi(f) = 0$, then $0 = q\overline{\varphi}[f] = [f]$ and $f \in I$. So $\mathrm{Ker}(\varphi) = I$ as well. From this we see that $G = (X_1 - g_1, \ldots, X_n - g_n)$ is a groebner basis for I.

- (Hilbert Polynomial).

  Definition 2.3.3. Let $I \subset \mathbb{F}[X_1, \ldots, X_n]$ be an ideal. We deffine the affine Hilbert function:
  $$ {}^a\mathrm{HF}_I : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0} : s \mapsto \dim_{\mathbb{F}}\left(\frac{\mathbb{F}[X_1, \ldots, X_n]_{\leq s}}{I_{\leq s}}\right). \tag{2.9} $$

  Example 2.3.4. For example, the number ${}^a\mathrm{HF}_{(0)}(s)$ is the number of n-tuples of nonnegative integers whose sum is $\leq s$. To compute this, write the generating function
  $$ \prod_{i=1}^{n} \frac{1}{1 - X_i} = \sum_{\alpha \in \mathbb{Z}^n_{\geq 0}} X^\alpha. \tag{2.10} $$

  Now put $X_i = t$, $\forall i$. We see that
  $$ (1-t)^{-n} = \sum_{s \geq 0} \binom{-n}{s} (-1)^s t^s = \sum_{s \geq 0} \binom{n+s}{s} t^s. \tag{2.11} $$

  So ${}^a\mathrm{HF}_{(0)}(s) = \binom{n+s}{s} = \frac{1}{n!}s^n + \cdots$.

  Example 2.3.5. Let $f \in \mathbb{F}[X_1, \ldots, X_n]$ be a degree d polynomial. Then
  $$ {}^a\mathrm{HF}_{(f)}(s) = \binom{s+n}{s} - \binom{s-d+n}{s-d} = \frac{d}{(n-1)!}s^{n-1} + \cdots. \tag{2.12} $$

An important property of the affine Hilbert function is the following:

Theorem 2.3.6. Let $I \subset \mathbb{F}[X_1, \ldots, X_n]$ be an ideal, then

$$ {}^a\mathrm{HF}_I(s) = {}^a\mathrm{HF}_{\mathrm{LT}(I)}(s),\ s \geq 0. \tag{2.13}$$

Another main result is the existence of the Hilbert polynomial.

Theorem 2.3.7. There exist a polynomial ${}^a\mathrm{HP}_I(t) \in \mathbb{Q}[t]$ such that

$$ {}^a\mathrm{HP}_I(s) = {}^a\mathrm{HF}_I(s),\ \forall s \geq 0. \tag{2.14}$$

Proof (sketch):

- Find a Groebner basis G of I
- Let $M \subset \mathbb{Z}_{\geq 0^n}$ be such that $\{X^\beta\}_{\beta \in M} = \{\mathrm{LM}(g) : g \in G\}$
- Write
$$ C(I) = \bigcap_{\beta \in M} C(\beta), C(\beta) = \{\alpha | X^\beta \nmid X^\alpha\} \tag{2.15}$$
as a union of $C(J, \tau) = \{\alpha | \alpha_j = \tau(j)\ \forall j \in J\}$. To do this use
$$ C(\beta) = \bigcup_{i=1}^{n} \bigcup_{t_i=0}^{\beta_i - 1} C(\{i\},\ \tau : i \mapsto t_i). \tag{2.16}$$
- Use the formula $|C(J, \tau)_{\leq s}| = \binom{d+s-|\tau|}{d}, d = n - |J|, |\tau| = \sum_{j \in J} \tau(j)$ and the inclusion-exclusion principle to compute
$$ {}^a\mathrm{HP}_I(s) = |C(I)_{\leq s}| = \left| \bigcup_{(J,\tau)} C(J, \tau)_{\leq s} \right|. \tag{2.17}$$

Example 2.3.8. Let $I = (X_1 - X_3^2, X_2 - X_3^3)$ and fix the LEX order with $X_1 > X_2 > X_3$. Then we calculate

- $G = \{X_1 - X_3^2, X_2 - X_3^3\}$
- $M = \{(1, 0, 0), (0, 1, 0)\}$
- We have
$$ \begin{aligned} C(I) &= C(\{1\}, \tau : 1 \mapsto 0) \cap C(\{2\}, \tau : 2 \mapsto 0) \\ &= C(\{1, 2\}, \tau : 1 \mapsto 0, 2 \mapsto 0). \end{aligned} \tag{2.18}$$

So we conclude that $|C(I)|_{\leq s} = \binom{1+s-0}{1} = s + 1$.

- (Algebraic dimension). Here we define the dimension of an algebraic set.

Definition 2.3.9. Let $V \subset \mathbb{F}^n$ be an algebraic set. We define its dimension as $\dim(V) = \deg {}^a\mathrm{HP}_{I(V)}(t)$. It has the property that it's the biggest integer $d \geq 1$ such that there exist indexes $1 \leq i_1 < \cdots < i_d \leq n$ such that $I(V) \cap \mathbb{F}[X_{i_1}, \ldots, X_{i_d}] = (0)$.

Example 2.3.10. Let $H_f = V(f)$, f irreducible, then

$$\begin{aligned} \dim(H_f) &= \deg {}^a\mathrm{HP}_{I(H_f)}(t) \\ &= \deg {}^a\mathrm{HP}_{(f)}(t) \\ &= \deg(\frac{\deg(f)}{(n-1)!}t^{n-1} + \cdots) \\ &= n-1. \end{aligned} \tag{2.19}$$

The dimension has the expected property: If $W \subset V$ is closed, then $\dim(W) \leq \dim(V)$. In fact

$$\begin{aligned} &I(V) \subset I(W) \Rightarrow \\ &I(V)_{\leq s} \subset I(W)_{\leq s} \Rightarrow \\ &{}^a\mathrm{HF}_{I(W)}(s) \leq {}^a\mathrm{HF}_{I(V)}(s),\ \forall s \geq 0 \Rightarrow \\ &\deg {}^a\mathrm{HF}_{I(W)}(t) \leq \deg {}^a\mathrm{HF}_{I(V)}(t). \end{aligned} \tag{2.20}$$

- (Transcendence degree). Let A be a field which is also a finitely generated $\mathbb{F}$-algebra. We define the transcendence degree of A over $\mathbb{F}$ (and write $\partial_{\mathbb{F}}(A)$) to be largest size of an algebraicaly independent subset of A over $\mathbb{F}$.

Example 2.3.11. We have $\partial_{\mathbb{F}}(\mathbb{F}[X_1, \ldots, X_n]) = n$. Indeed, let $\{f_1, \ldots, f_r\}$ be polynomials and suppose that $r > n$. Write

$$\begin{aligned} &W_s = \{f^\beta : \beta \in \mathbb{Z}^r_{\geq 0},\ |\beta| \leq s\},\ d = \max\{\deg(f_i) : i = 1, \ldots, r\} \\ &V_s = \{X^\alpha : |\alpha| \leq s\} \end{aligned} \tag{2.21}$$

Then $W_s \subset V_{ds}$ and

$$|W_s| = \binom{r+s}{s} \overset{s \gg 0}{>} \binom{n+sd}{sd} = \dim(V_{sd}) \geq \dim(W_s). \tag{2.22}$$

So, for $s \gg 0$ there exist $\{a_\beta\}_\beta \subset \mathbb{F}$ such that $\sum_\beta a_\beta f^\beta = 0$ and $\{f_1, \ldots, f_r\}$ is algebraicaly dependent.

Theorem 2.3.12. Let V be an algebraic set and write $I = I(V)$, then

$$\dim(V) = \partial_{\mathbb{F}}(A[V]) = \partial_{\mathbb{F}}(\frac{\mathbb{F}[X_1, \ldots, X_n]}{I}). \tag{2.23}$$

Proof: Let $d = \dim(V)$, so there exist indexes $1 \leq i_1 < \cdots < i_d \leq n$ such that $I \cap \mathbb{F}[X_{i_1}, \ldots, X_{i_d}] = (0)$. This is equivalent to say that $[X_{i_1}], \ldots, [X_{i_d}] \in A[V]$ are algebraicaly independent over $\mathbb{F}$. Therefore $d \leq \partial_{\mathbb{F}}(A[V])$. Now, suppose that $\varphi_i = [f_i] \in A[V], i = 1, \ldots m$ are algebraicaly independent over $\mathbb{F}$ and put $N = \max \deg(f_i)$, then the map

$$\mathbb{F}[Y_1, \ldots, Y_m]_{\leq s} \to \frac{\mathbb{F}[X_1, \ldots, X_n]_{\leq Ns}}{I_{\leq Ns}} : g \mapsto g(\varphi_1, \ldots, \varphi_m) \tag{2.24}$$

is injective. Then, by taking dimensions we get

$$\binom{m+s}{s} \leq {}^a\mathrm{HF}_I(s), \ \forall s \geq 0. \tag{2.25}$$

We know that both expressions are polynomials in s, so by taking degrees we conclude that $m \leq \dim(V) = d$ and $\partial_{\mathbb{F}}(A[V]) \leq d$. □

With small changes in the above proof, we can verify that in the case of $A[V]$ being an integral domain, we have also $\partial_{\mathbb{F}}(K) = \dim(V), \ K = \mathrm{Frac}(A[V])$.

Example 2.3.13. Let $C = V(X_1 - X_3^2, X_1 - X_3^3)$, then $I(C) = (X_1 - X_3^2, X_2 - X_3^3)$ and

$$\frac{\mathbb{F}[X_1, X_2, X_3]}{I(C)} \cong \mathbb{F}[X_3] \tag{2.26}$$

so $\partial_{\mathbb{F}}(A[C]) = 1$.

Example 2.3.14. The purpose of this example is to show that if $V \subset \mathbb{F}^n, |\mathbb{F}| = \infty$ is a linear space, then $\dim(V) = \dim_{\mathbb{F}}(V)$.

- Let $\mathrm{Ann}(V) = \mathrm{Span}(f_1, \ldots, f_m)$. Since V has finite dimension, we have $v \in V \iff f(v) = 0 \ \forall f \in \mathrm{Ann}(V)$ and $V = V(\{f_1, \ldots, f_m\})$.
- Let $f_i = \sum a_{ij} X_j$, where we identify X_j with the jth coordinate functions in $\mathbb{F}^n$. Now, if $A = (a_{ij})$ and apply reduction to put A in the form:

$$\begin{pmatrix} a_{1j_1} & * & 0 & * & \cdots & 0 & \cdots \\ & & a_{2j_2} & * & \cdots & 0 & \cdots \\ & & & & \vdots & \cdots & \vdots \\ & & & 0 & \cdots & a_{mj_m} & \cdots \end{pmatrix} \tag{2.27}$$

  with $a_{ij_i} = 1, \ j_1 < j_2 < \cdots$. Now put LEX with: $X_{j_1} > \cdots X_{j_m} > X_j \ j \notin \{j_1, \ldots, j_m\}$ and write $f_i = X_{j_i} + h_i$ with $h_i \in \mathbb{F}[X_j : j \notin \{j_1, \ldots, j_m\}]$. Then we see that $A[V] \cong \mathbb{F}[X_j : j \notin \{j_1, \ldots, j_m\}]$ and $(f_1, \ldots, f_m)$ is prime. Therefore $I(V) = (f_1, \ldots, f_m)$. Finally

$$\dim(V) = \partial_{\mathbb{F}}(A[V]) = \dim_{\mathbb{F}} \mathbb{F}[X_j : j \notin \{j_1, \ldots, j_m\}] = \dim_{\mathbb{F}} V. \tag{2.28}$$

2.4 Tangent Spaces

- (Regular maps). Let $V \subset \mathbb{F}^m, W \subset \mathbb{F}^n$ be algebraic sets. Then a map $f : V \to W$ is said to be regular if there exist polynomials such that $f = (f_1, \ldots, f_m)$. Such a map is continuous in the Zariski topology.

  Remark 2.4.1. We can identify $A[V]$ with the set of regular functions $V \to \mathbb{F}$. In this context, we have a contravariant functor:

  $$\varphi : V \to W \rightsquigarrow \varphi^* : A[W] \to A[V] : [g] \mapsto [g] \circ \varphi \tag{2.29}$$

  Example 2.4.2. Let $\varphi : \mathbb{F} \to C : x \mapsto (x^3, x^2, x)$ and write $C = (X_1 - X_3^2, X_2 - X_3^3)$. Remember the isomorphism $A[C] \to \mathbb{F}[Y] : [X_1] \mapsto Y^2, [X_2] \mapsto Y^3, [X_3] \mapsto Y$. So we have

  $$\varphi^* : A[C] \to A[\mathbb{F}] : [X_3] \mapsto [X_3] \circ \varphi. \tag{2.30}$$

  The first map is $[X_3] : C \to \mathbb{F} : (x^3, x^2, x) \mapsto x$ and the last map is $\mathbb{F} \to C : x \mapsto x$. At the algebraic level, the map φ^* is just the the isomorphism $\mathbb{F}[Y] \cong \mathbb{F}[X]$.

- (Tangent spaces). Let V be an algebraic set and $k_p = \mathbb{F}$ be the $A[V]-$module with action $[f]\alpha \equiv f(p)\alpha$. Let $I(V) = (f_1, \ldots, f_m)$. We define

  $$\begin{aligned} T_p(V) &= \{v \in \mathbb{F}^n : d(f)_p(v) = 0 \ \forall f \in I(V)\} \\ &= \{v \in \mathbb{F}^n : d(f_i)_p(v) = 0 \ \forall i\} \\ &= V(\{d(f_i)_p : i = 1, \ldots, m\}) \end{aligned} \tag{2.31}$$

  Remark 2.4.3. Caution ! The polynomials of f_i here are not the generators of V, but of $I(V)$. For example, if $V = V(f)$, we can conlude that $T_p(V) = V(df_p)$ only if f is irreducible, i.e., if $I(V) = (f)$.

  Here $df_p \equiv \sum_{i=1}^n D_i(f)(p)X_i \in \mathbb{F}[X_1, \ldots, X_n]$. We have an isomorphism

  $$T_p(V) \to \mathrm{Der}_{\mathbb{F}}(A[V], k_p) : v \mapsto D_v;\ D_v([f]) = \sum_{i=1}^n [D_i(f)] \cdot v_i \in \mathbb{F}. \tag{2.32}$$

  with inverse $D \mapsto (D([X_i]))_{i=1}^n$. Of course $D_v([f]) = df_p(v)$.

  Example 2.4.4. Let $C = (f_1, f_2),\ f_1 = X_1 - X_3^2, f_2 = X_2 - X_3^3$ and $p = (x^3, x^2, x) \in C$. We have

  $$d(f_1)_p = 1 - 2xX_3,\ d(f_2) = 1 - 3x^2X_3 \tag{2.33}$$

  and $T_p(C) = \mathbb{F}\langle(3x^2, 2x, 1)\rangle$.

  Example 2.4.5. The determinant is an irreducible polynomial. Indeed, let $\det \in \mathbb{F}[X_{ij} : 1 \leq i, j \leq n]$. Then, if we apply Laplace theorem on the first row, we get

  $$\det = A_{11}X_{11} - A_{12}X_{12} + \cdots + (-1)^n A_{1n}X_n \tag{2.34}$$

where A_{ij} denotes the determinant of the matrix obtained by excluding the ith row and jth collumn. By induction on the number of variables, we assume that A_{ij} are irreducible polynomials. Now, suppose that $\det = fg$, where $\deg(f), \deg(g) > 0$. Choose LEX with $X_{ij} > X_{k\ell}$ if $i < k$ or if $i = k$ and $j < \ell$. Then

$$\mathrm{LT}(A_{11})X_{11} = \mathrm{LT}(\det) = \mathrm{LT}(f)\mathrm{LT}(g). \tag{2.35}$$

Assume that $X_{11}|\mathrm{LT}(f)$ (so $\mathrm{LT}(g) < X_{11}$) and write $f = AX_{11} + B$, $\mathrm{LT}(B) < X_{11}$. Then we must have $AgX_{11} = A_{11}X_{11}$ and $A_{11} = Ag$. Since $\deg(g) > 1$ and A_{11} is irreducible, we must have $g = A_{11}$ and $A = 1$. Now we get

$$-A_{12}X_{12} + \cdots + (-1)^n A_{1n}X_n = BA_{11} \tag{2.36}$$

and $-\mathrm{LT}(A_{12})X_{12} = \mathrm{LT}(B)\mathrm{LT}(A_{11})$. Since $X_{22}|\mathrm{LT}(A_{11})$, we are led to $X_{22}|\mathrm{LT}(A_{12})$ which is a contradiction because X_{22} doenst appear in A_{12} by definition.

Now, since det is an irreducible homogeneous polynomial, the difference $\det - 1$ must be irreducible. Therefore $I(SL_n\mathbb{F}) = (\det - 1)$ and we can compute the tangent space $T_I(SL_n) = V(d_I(\det))$. We have

$$\begin{aligned}
d_I \det &= \sum_{k,\ell} D_{k\ell}(\det)(I)X_{k\ell} \\
&= \sum_{k,\ell} \sum_{\sigma \in S_n} D_{k\ell} \prod_{i=1}^{n} X_{i\sigma(i)}(I)X_{k\ell} \\
&= \sum_{k,\ell} \sum_{\sigma \in S_n} \delta_{\ell,\sigma(k)} \prod_{i \neq k} \delta_{i,\sigma(i)} X_{k\ell} \\
&= \sum_{k,\ell} \delta_{k\ell} X_{k\ell} \quad (\delta_{\ell,\sigma(k)} \prod_{i \neq k} \delta_{i,\sigma(i)} \neq 0 \text{ only if } \sigma = 1) \\
&= \sum_{k} X_{kk}.
\end{aligned} \tag{2.37}$$

Therefore $T_I SL_n(\mathbb{F}) = \mathfrak{sl}_n\mathbb{F}$, the Lie algebra of traceless matrices.

- (Gauss lemma).

Remark 2.4.6. Remember that irreducible elements in a UFD are prime. This and the Gauss lemma below, gives a good way to test if a polynomial generates a prime ideal.

We state the Gauss lemma.

Lemma 2.4.7. Let R be a UFD ring (think in $\mathbb{F}[X]$), then $R[X]$ is a UFD.

Proof: Let f and $K = \mathrm{Frac}(R)$. Since $K[X]$ is a UFD, we have $f = FG$, $F, G \in K[X]$ with F irreducible and $\deg(G) < \deg(f)$. Since R is a UFD, there exist $r, s \in R$

such that $rF, sG \in R[X]$. Decompose $rs = t_1 \cdots t_k$ into irreducible elements. Then if $F' = rF, G' = sG$, we have

$$[F'][G'] = [rs][f] = 0 \in R/(t_i)[X] \tag{2.38}$$

because $f \in R[X]$. So, since $R/(t_i)[X]$ is a domain, we see that $t_i | F'$ or $t_i | G'$. Then we see that there exist a partition $\{1, \ldots, k\} = I \cup J$ such that $F' / \prod_{i \in I} t_i, G' / \prod_{j \in J} t_j \in R[X]$. It's clear that F' still irreducible, then an induction on degree yields the result. □

Corollary 2.4.8. Let $f, g \in R = \mathbb{F}[X_1, \ldots, X_n]$ and assume f irreducible with $f \nmid g$, then there exist $F, G \in R$ such that

$$Gf + Fg = h \in \mathbb{F}[X_1, \ldots, X_{n-1}] \setminus 0. \tag{2.39}$$

Proof: Let $K = \mathrm{Frac}(\mathbb{F}[X_1, \ldots, X_{n-1}])$, then $f, g \in K[X_n]$ and there exist $A, B \in K[X_n]$ such that $Af + Bg = 1$ (by Gauss lemma f still irreducible as an element of $K[X_n]$). Then there exist $h \in \mathbb{F}[X_1, \ldots, X_{n-1}]$ such that $hA, hB \in \mathbb{F}[X_1, \ldots, X_{n-1}]$. So by writing $G = hA, F = hB$ we are done. □

Notation 2.4.9. Let M be a $A-$module (where A is a k-algebra), then we define

$$\mathrm{Der}_k(A, M) = \{D \in \mathrm{Hom}_k(A, M) \mid D(xy) = x \cdot D(y) + y \cdot D(x),\ \forall x, y \in A\}. \tag{2.40}$$

Observe that since M is a $A-$module, it's also a $k-$vector space with action given by $\alpha \cdot m \equiv (\alpha 1_A) \cdot m, \alpha \in k$.

Notation 2.4.10. Let $A = \mathbb{F}[X_1, \ldots, X_n]/I$, where M is a $A-$module. We write

$$\mathcal{T}_{A,M} = \{(v_1, \ldots, v_n) \in M^n : \sum_i [D_i(f)] \cdot v_i = 0,\ \forall f \in I\}. \tag{2.41}$$

We have the isomorphism $\mathcal{T}_{A,M} \to \mathrm{Der}_{\mathbb{F}}(A, M) : v \mapsto D_v$ with inverse

$$D \mapsto (D(\overline{X_i}))_{i=1}^n. \tag{2.42}$$

Now consider the following context, let $K = \mathrm{Frac}(A)$ as A-module (A acts by multiplication). Then we have an isomorphism

$$\mathrm{Der}_{\mathbb{F}}(K, K) \to \mathrm{Der}_{\mathbb{F}}(A, K) : D \mapsto D|_A. \tag{2.43}$$

In particular we also have the isomorphism $\mathcal{T}_{A,K} \cong \mathrm{Der}_{\mathbb{F}}(K, K)$. We observe that this is also a $K-$vector space isomorphism, it's a bijection compatible with the $K-$action. From this isomorphism, we see that $\mathcal{T}_{A,K}$ only depends on K, and the fact that $\mathrm{Frac}(A) = K$.

Theorem 2.4.11. Let $A = \mathbb{F}[X_1, \ldots, X_n]/I$ be a domain and $K = \mathrm{Frac}(A)$. Then

$$\partial_{\mathbb{F}}(A) = \dim_K \mathrm{Der}_{\mathbb{F}}(K, K). \tag{2.44}$$

Proof: Let $d = \partial_{\mathbb{F}}(A) = \partial_{\mathbb{F}}(K)$, then there exist $z_1, \ldots, z_{d+1}$ such that

- $K = \mathbb{F}(z_1, \ldots, z_{d+1})$
- $\{z_1, \ldots, z_d\}$ is algebraicaly independent
- z_d is separable over $\mathbb{F}(z_1, \ldots, z_d)$

Let $R = \mathbb{F}[z_1, \ldots, z_{d+1}] \subset K$ and $f \neq 0$ irreducible such that $f(z_1, \ldots, z_{d+1}) = 0$. Then by corollary (2.4.8), we see that $\alpha : K[Y_1, \ldots, Y_{d+1}] \to K : g \mapsto g(z_1, \ldots, z_{d+1})$ has kernel (f). Therefore

$$\begin{aligned} \mathcal{T}_{R,K} &= \{(v_1, \ldots, v_{d+1}) \in K^{d+1} : \sum_i [D_i(f)] \cdot v_i = 0\} \\ &= \{(v_1, \ldots, v_{d+1}) \in K^{d+1} : \sum_i D_i(f)(z_1, \ldots, z_{d+1}) \cdot v_i = 0\}. \end{aligned} \tag{2.45}$$

We see that $D_i(f)(z_1, \ldots, z_{d+1}) = 0 \Rightarrow D_i(f) \in (f)$ and $D_i(f) = 0$. Since $f \neq 0$ and $|\mathbb{F}| = \infty$, we must have $D_i(f) \neq 0$ for some i, so $D_i(f)(z_1, \ldots, z_{d+1}) \neq 0$ for some i. Then we conclude that

$$\dim_K \mathrm{Der}_{\mathbb{F}}(K, K) = \dim_K \mathcal{T}_{R,K} = d = \partial_{\mathbb{F}}(K) = \partial_{\mathbb{F}}(A). \tag{2.46}$$

□

Corollary 2.4.12. If $A = \mathbb{F}[X_1, \ldots, X_n]/(f_1, \ldots, f_m)$, then

$$\partial_{\mathbb{F}}(A) = \dim_K \mathcal{T}_{A,K} = n - \mathrm{Rank}_K([D_i(f_j)]). \tag{2.47}$$

Example 2.4.13. Let $C = V(f_1, f_2), f_1 = X_1 - X_3^2, f_2 = X_2 - X_3^3$. We have

$$A[C] = \frac{\mathbb{F}[X_1, X_2, X_3]}{(f_1, f_2)} \to \mathbb{F}[X_3] : [g] \mapsto g(X_3^3, X_3^2, X_3). \tag{2.48}$$

So $A[C]$ is a domain and we can define $K = \mathrm{Frac}(A[C])$. We compute

$$\mathrm{Rank}_K([D_i(f_j)]) = \mathrm{Rank}_K \begin{pmatrix} 1 & 0 & -2[X_3] \\ 0 & 1 & -3[X_3] \end{pmatrix} = 2. \tag{2.49}$$

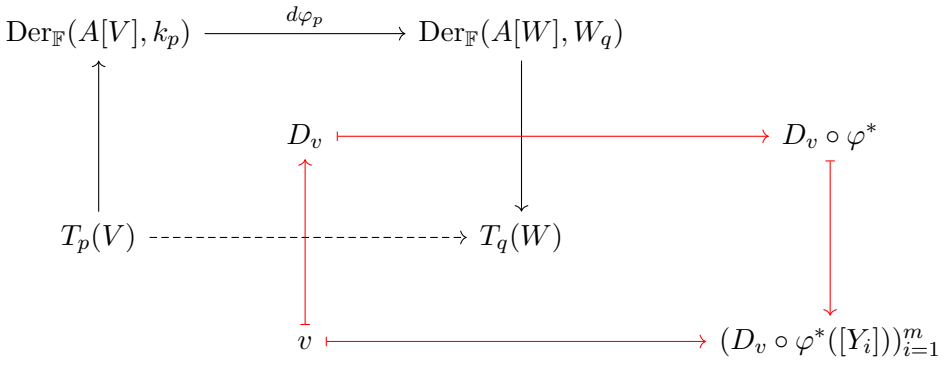
Then we conclude $\dim(V) = 3 - 2 = 1$.

- (differential of a regular map). Here we define the notion of differential of a regular map.

Definition 2.4.14. Let $\varphi : V \to W$ be a regular map between algebraic varieties. Also, let $p \in V, q = \varphi(p) \in W$. We define:

$$d\varphi_p : \mathrm{Der}_{\mathbb{F}}(A[V], k_p) \to \mathrm{Der}_{\mathbb{F}}(A[W], k_q) : D_v \mapsto D_v \circ \varphi^* \tag{2.50}$$

At the level of tangent spaces, we have:



In more details, if we write $\varphi = (f_1, \ldots, f_m)$, then $\varphi^*([Y_i]) = [f_i]$ and we get

$$D_v([f_i]) = \sum_{j=1}^{m} [D_j(f_i)] \cdot v_j = \sum_{j=1}^{m} D_j(f_i)(p) v_j. \tag{2.51}$$

Therefore

$$d\varphi_p(v) = (D_j(f_i)(p))_{i,j} \cdot v = J_p(\varphi) \cdot v. \tag{2.52}$$

Example 2.4.15. Let $G \subset \mathbb{F}^{n^2}$ be an algebraic group and $\mu : G \times G \to G$ be the product of matrices. We have $G \times G \subset \mathbb{F}^{n^2} \times \mathbb{F}^{n^2} \cong \mathbb{F}^{2n^2}$. Now, identify the first n^2 coordinate functions with X_{ij} and the second n^2 coordinate functions with Y_{ij}. So we have

$$\mu_{ij} = \sum_{\ell} X_{i\ell} Y_{\ell j}. \tag{2.53}$$

Then we have

$$\frac{\partial \mu_{ij}}{\partial X_{rs}}(I) = \frac{\partial \mu_{ij}}{\partial Y_{rs}}(I) = \delta_{ir}\delta_{sj}. \tag{2.54}$$

Therefore

$$d_{(I,I)}\mu(A, B) = (\sum_{r,s} \frac{\partial \mu_{ij}}{\partial X_{rs}}(I) A_{rs} + \frac{\partial \mu_{ij}}{\partial Y_{rs}}(I) B_{rs})_{ij} = (A_{ij} + B_{ij}) = A + B. \tag{2.55}$$

2.5 Notes from Springer

From now on, we will follow the book ([4]).

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